ECON 21031 (S26)
Econometrics II — Honors

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1 Linear Regression Mechanics

1.1 Three Interpretations and Two Consequences

Three Interpretations

(i) Model: assume $\mu(x) := \mathbb{E}[Y_i|X_i = x] = x'\beta$ for some β.

(ii) Best linear predictor: $X_i'\beta$ is the best linear predictor of Y_i given X_i.

(iii) Best linear approximation: $X_i'\beta$ is the best linear approximation of the conditional mean μ.

Each of these leads to

$$\beta = \mathbb{E}[X_i X_i']^{-1}\,\mathbb{E}[X_i Y_i].$$

When using interpretations (ii) or (iii), one should keep in mind that the word "best" is only true in a very weak sense. It is contingent on fixing the choice of x.

The empirical analogue of β is

$$b = \operatorname*{arg\,min}_{b\in\mathbb{R}^k} \frac{1}{n}\sum_{i=1}^{n}(y_i - x_i'b)^2 = \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} x_i y_i\right).$$

Define the **population residuals** as $U_i := Y_i - X_i'\beta$.

Two Consequences With the model interpretation, we have **mean independence**:

$$\mathbb{E}[U_i \mid X_i] = \mathbb{E}[Y_i \mid X_i] - X_i'\beta = 0.$$

The two other interpretations gives the weaker **orthogonality**:

$$\mathbb{E}[X_i U_i] = \mathbb{E}[X_i Y_i] - \mathbb{E}[X_i X_i']\beta = 0.$$

Orthogonality holds also for the OLS estimate and residuals (replace all $\mathbb{E}$ with the sample expectation $\mathbb{E}_n$).

In particular, if there is a constant regressor say X_{ik}, then $\mathbb{E}[U_i \times 1] = 0$, so $\mathrm{Cov}(U_i, X_{ik}) = 0$. Note that mean independence implies orthogonality, but not vice versa.

1.2 Short and Long Regressions

Suppose we regress Y_i on X_{i1}. How do the coefficients on X_{i1} change if we add X_{i2} as a regressor? The coefficient on X_{i1} in the short regression is

$$\begin{aligned}\beta_{1s} &= \mathbb{E}[X_{i1}X_{i1}']^{-1}\,\mathbb{E}\big[X_{i1}\underbrace{(X_{i1}'\beta_{1l} + X_{i2}'\beta_{2l} + U_i)}_{Y_i \text{ in long regression}}\big]\\ &= \beta_{1l} + \mathbb{E}[X_{i1}X_{i1}']^{-1}\,\mathbb{E}[X_{i1}X_{i2}']\beta_{2l} =: \beta_{1l} + \alpha\beta_{2l}.\end{aligned}$$

where α is a $d_1 \times d_2$ matrix of coefficients from regressing X_{i2} on X_{i1}; the jth column of α is the coefficient from regressing X_{i2j} on X_{i1}. This is often called the **omitted**

variable bias formula — but note that this label presupposes that the long regression is the true model.

Note that the short and long regression coefficients are the same if either of the following holds:

(i) $\beta_{2l} = 0$, i.e., X_{i2} is not a predictor of Y_i after controlling for X_{i1}.
(ii) $\alpha = 0$, i.e., X_{i1} and X_{i2} are orthogonal.

1.3 The Frisch-Waugh-Lovell Theorem

Consider

$$Y_i = X_{i1}'\beta_1 + X_{i2}'\beta_2 + U_i.$$

We want an expression for β_1 that is easy to work with. The idea is to regress X_{i1} on X_{i2},

$$X_{i1} = \gamma' X_{i2} + V_i,$$

and then replace X_{i1} with V_i. Since all changes can be "captured" by shifting coefficients on X_{i2}, this procedure should not change the coefficients on X_{i1}. Moreover, V_i is orthogonal to X_{i2}, so we can ignore X_{i2} when regressing Y_i on V_i (case (ii) above).

Thus we have

Theorem 1.1 (Frisch-Waugh-Lovell, FWL)**.** *Consider regressing Y_i on $X_i = [X_{i1}', X_{i2}']$, with coefficients $\beta = (\beta_1', \beta_2')'$. If $V_i = [V_{i1}, \ldots, V_{id_1}]'$ is the vector of residuals from regressing each X_{i1j} on all of X_{i2}, then*

$$\beta_1 = \mathbb{E}[V_i V_i']^{-1}\,\mathbb{E}[V_i Y_i].$$

That is, β_1 is determined by the variation in X_{i1} left-over after regressing on X_{i2}.

Proof. Note that

$$Y_i = \beta_1'(\gamma' X_{i2} + V_i) + X_{i2}'\beta_2 + U_i = \beta_1' V_i + X_{i2}'(\gamma\beta_1 + \beta_2) + U_i.$$

Since V_i and X_{i2} are both orthogonal to U_i, the coefficients in the long regression are just β_1 and $\gamma\beta_1 + \beta_2$. Since V_i is orthogonal to X_{i2}, the coefficient in the short regression is also β_1. □

Remark 1.2 (Double Residual Regression). Note that there is no need to residualize Y_i for this statement of FWL. The reason we sometimes also residualize Y_i is that the residuals from the **residual regression** $Y_i = V_i'\beta_1 +$ (residual) are not the same as the residuals of the long regression $Y_i = X_{i1}'\beta_1 + X_{i2}'\beta_2 + U_i$. However, a **double residual regression** of the residuals of Y_i on the residuals of X_{i1} gives the same coefficients and residuals as the long regression.

To see this, note that from orthogonality we have the residuals of a regression of Y_i on X_{i2} is

$$R_i = Y_i - X_{i2}'(\gamma\beta_1 + \beta_2) = \beta_1' V_i + U_i.$$

Thus the residuals of the regression of R_i on V_i is precisely the original residuals U_i. ☕

1.3.1 Applications of FWL

Constant. Suppose X_{i2} is the constant, and X_{i1} are all other regressors. The residuals of regressing X_{i1} on X_{i2} are then the de-meaned regressors: $V_i = X_{i1} - \mathbb{E}[X_{i1}]$. Using this, one can show that the coefficients on X_{i1} are

$$\beta_1 = \mathbb{E}[V_i V_i']^{-1}\, \mathbb{E}[V_i Y_i] = \operatorname{Var}[X_{i1}]^{-1} \operatorname{Cov}(X_{i1}, Y_i).$$

This is the vector-matrix analogue of the formula for the slope coefficient in a simple regression. From this we see that β_1 does not change if we add a constant term to X_{i1}.

Fixed Effects. Let X_{i2} be a vector of **fixed effects**[1] for a categorical variables W_i. Let X_{i1} be other variables. Let V_i denote the residuals from regressing X_{i1} onto the fixed effects. This is a saturated regression, so $V_i = X_{i1} - \mathbb{E}[X_{i1}|X_{i2}]$ are the within-group de-meaned regressors.

FWL says that the coefficients on X_{i1} in the regression with fixed effects are

$$\beta_1 = \mathbb{E}[V_i V_i']^{-1}\, \mathbb{E}[V_i Y_i].$$

To interpret, note that

$$\begin{aligned}
\mathbb{E}[V_i Y_i] &= \mathbb{E}\left[(X_{i1} - \mathbb{E}[X_{i1}|X_{i2}])Y_i\right] \\
&= \mathbb{E}\Big[\, \mathbb{E}\left[(X_{i1} - \mathbb{E}[X_{i1}|X_{i2}])Y_i|X_{i2}\right] \Big] \\
&= \mathbb{E}\left[\operatorname{Cov}(X_{i1}, Y_i|X_{i2})\right] = \mathbb{E}[\beta_1(X_{i2}) \operatorname{Var}(X_{i1}|X_{i2})],
\end{aligned}$$

where β_1 is the within-bin coefficients. The last equality follows from $\beta_1(X_2) = \operatorname{Var}(X_{i1}|X_{i2})^{-1} \operatorname{Cov}(X_{i1}, Y_i|X_{i2})$. Also,

$$\begin{aligned}
\mathbb{E}[V_i V_i'] &= \mathbb{E}\left[\mathbb{E}\left[(X_{i1} - \mathbb{E}[X_{i1}|X_{i2}])(X_{i1} - \mathbb{E}[X_{i1}|X_{i2}])'|X_{i2}\right]\right] \\
&= \mathbb{E}\left[\operatorname{Var}(X_{i1}|X_{i2})\right].
\end{aligned}$$

Thus, in a model with fixed effects, the coefficients β_1 are the *normalized weighted average of the within-bin coefficients*, with weights proportional to the within-bin variation in X_{i1}. One can read from the formula that β_1 is unchanged if we add to X_{i1} any function of X_{i2}.

Also, note that more weights are put on bins with more variation in X_{i1}, which generate more informative signal and lowers the overall variance of the estimator — this is just a consequence of OLS being the BLUE. For purpose of interpretation, it is preferred of course to (manually) weight the within-bin coefficients by the respective bin sizes instead.

Computational Considers of Fixed Effects. Assume the same setup as above. To compute the coefficients on X_{i1} in the regression with fixed effects, one can first de-mean X_{i1} and Y_i by the fixed effects, and then regress the de-meaned Y_i on the de-meaned X_{i1}.

[1] I.e., dummies for categorical variables. Though one would not use this term when the possible values are very few, e.g., a variable for sex.

When X_{i2} can take on many values, this is more computationally efficient than having a set of dummies.

With multiple fixed effects this is more complicated but can be done when the fixed effects have no interactions, using what's called the **alternating projections algorithm**. Consider say two fixed effects W_i and Z_i. This algorithm iteratively de-means X_{i1} from W_i, then from Z_i, then from W_i, and so on until convergence.

1.4 The Geometry of Least Squares

OLS is a **projection** of y onto $\operatorname{col} x$:

$$b = \arg\min_b \|y - xb\|^2.$$

We have

$$\hat{y} := xb = x\left(\frac{1}{n}\sum x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum x_i y_i\right) = x(x'x)^{-1}x'y,$$

where $p := x(x'x)^{-1}x'$ is the **projection matrix**.

Proposition 1.3. *A matrix is idempotent and symmetric if and only if it is an orthogonal projection matrix.*

Symmetry is what makes a projection an *orthogonal* projection.

The matrix $m := \mathrm{Id} - p$ is called the **residual-maker matrix** (or the annihilator matrix) associated with x. It is an orthogonal projection onto the orthogonal complement of $\operatorname{col} x$. Thus we can interpret OLS as an orthogonal decomposition:

$$y = py + my.$$

This gives $\|py\|, \|my\| \leq \|y\|$.

With a constant term so that $\sum \hat{u} = 0$, we have in the sample that $\|y - \bar{y}\|^2 = \|\hat{y} - \bar{y}\|^2 + \|\hat{u} - 0\|^2$. This naturally motivates the definition of the R^2 as

$$R^2 = \frac{\|\hat{y} - \bar{y}\|}{\|y - \bar{y}\|} = 1 - \frac{\|\hat{u}\|}{\|y - \bar{y}\|}.$$

1.4.1 The Geometry of Long and Short Regressions

Recall that by FWL in the sample,

$$b_{1s} = b_{1l} + (x_1'x_1)^{-1}(x_1'x_2)b_{2l} =: b_{1l} + ab_{2l}$$

The two cases when short and long coefficients coincide are:

1. $b_{2l} = 0$: long coefficients are zero. That is, $\hat{y}_l$ is already in $\operatorname{col} x_1$, so removing x_2 doesn't change the fitted values.
2. $a = 0$: x_1 and x_2 are orthogonal. That is, the closest point to y in $\operatorname{col} x_1$ is still the closest when we can add $\operatorname{col} x_2$.

1.4.2 The Geometry of FWL

Geometrically, $x_1 = p_2 x_1 + m_2 x_1$ takes $\operatorname{col} x_1$ and rotates/reparameterizes it so that it is contained in $\operatorname{col}(x_2)^\perp$. The first component of the projection onto $\operatorname{col} x_1 + \operatorname{col} x_2$ (β_1 in the long regression) is precisely the projection onto reparameterized $\operatorname{col} x_1$ subspace (coefficient of the residual regression).

1.4.3 Geometry in the Population

The same geometry holds for random variables in the space L^2. The analogues of the projection and residual-maker matrices are the linear operators

$$\mathbb{L}[Y_i|X_i] \coloneqq X_i' \,\mathbb{E}[X_i X_i']^{-1}\,\mathbb{E}[X_i Y_i], \quad \mathbb{R}[Y_i|X_i] \coloneqq Y_i - \mathbb{L}[Y_i|X_i].$$

The operators $\mathbb{L}$ and $\mathbb{R}$ are linear, idempotent, and self-adjoint (this is the analogue of symmetry for matrices):

$$\mathbb{E}\left[\mathbb{L}[Y_i|X_i] Z_i'\right] = \mathbb{E}\left[Y_i \mathbb{L}[Z_i|X_i]'\right].$$

One can show this simply by expanding the definition.

2 Linear Regression Statistics

2.1 Unbiasedness, Consistency, and Asymptotic Normality

Recall the OLS estimator can be written as

$$B = \beta + \left(\frac{1}{n}\sum X_i X_i'\right)^{-1}\left(\frac{1}{n}\sum X_i U_i\right)$$

Thus the bias is

$$\mathbb{E}[B - \beta|X] = \left(\frac{1}{n}\sum X_i X_i'\right)^{-1}\left(\frac{1}{n}\sum X_i \,\mathbb{E}[U_i|X]\right).$$

Proposition 2.1. *If U_i are independent over i and $\mathbb{E}[Y|X = x] = x'\beta$ is linear, then B is an unbiased estimator of β.*

Proof. Independence gives $\mathbb{E}[U_i|X] = \mathbb{E}[U_i|X_i]$ and linearity gives $\mathbb{E}[U_i|X_i] = \mathbb{E}[Y_i|X_i] - X_i'\beta = 0$. $\square$

The WLLN implies that

$$\frac{1}{n}\sum X_i X_i' \xrightarrow{\text{p}} \mathbb{E}[X_i X_i'], \quad \frac{1}{n}\sum X_i U_i \xrightarrow{\text{p}} \mathbb{E}[X_i U_i] = 0.$$

Thus the continuous mapping theorem gives

Proposition 2.2. *The OLS estimator B is consistent for β.*

Write

$$\sqrt{n}(B - \beta) = \left(\frac{1}{n}\sum X_i X_i'\right)^{-1}\left(\frac{1}{\sqrt{n}}\sum X_i U_i\right).$$

CLT, WLLN, and Slutsky gives

Proposition 2.3.

$$\sqrt{n}(B - \beta) \xrightarrow{\text{d}} \mathcal{N}\left(0, \mathbb{E}[X_i X_i']^{-1}\,\mathbb{E}[U_i^2 X_i X_i']\,\mathbb{E}[X_i X_i']^{-1}\right) =: \mathcal{N}(0, \Sigma).$$

Using the sample analogue, we construct the following estimator for Σ:

$$\hat{\Sigma} = \left(\frac{1}{n}\sum X_i X_i'\right)^{-1}\left(\frac{1}{n}\sum \hat{U}_i^2 X_i X_i'\right)\left(\frac{1}{n}\sum X_i X_i'\right)^{-1}.$$

We can again use WLLN and CMP to show that $\hat{\Sigma} \xrightarrow{\text{p}} \Sigma$.

2.1.1 Homoskedasticity

Under the **homoskedasticity** assumption that $\mathbb{E}[U_i^2|X_i] = \mathbb{E}[U_i^2]$, the asymptotic variance simplifies to

$$V = \mathbb{E}[U_i^2]\,\mathbb{E}[X_i X_i']^{-1} =: V_{\text{HOM}}.$$

2.1.2 Studentization

Write $S \coloneqq \hat{\Sigma}^{1/2}$. Then $S \xrightarrow{p} \Sigma^{1/2}$ and so

$$(S/\sqrt{n})^{-1}(B-\beta) \xrightarrow{d} \mathcal{N}(0, \mathrm{Id}).$$

2.1.3 F-Test

The **Wald statistic** for $H_0 : r\beta = q$ is

$$\|\sqrt{n}V_r^{-1/2}(rB-q)\|^2 \xrightarrow{d} \chi^2_{d_r}, \quad V_r \coloneqq rVr'.$$

The F-statistic for the same test is the Wald statistic divided by d_r.

In the context of an OLS, the overall F**-test** refers to the test that all components of β (except for the constant coefficient) are zero.

2.2 Variance Mechanics

2.2.1 Signal and Noise

For the simple linear regression with $X_i = [X_{i1}, 1]'$ recall

$$\mathbb{E}[X_i X_i']^{-1} = \frac{1}{\mathrm{Var}[X_{i1}]}\begin{bmatrix} 1 & -\mathbb{E}[X_{i1}] \\ -\mathbb{E}[X_{i1}] & \mathbb{E}[X_{i1}^2] \end{bmatrix}$$

Assuming homoskedasticity, we have[2]

$$\Sigma = \Sigma_{\mathrm{HOM}} = \mathbb{E}[U_i^2]\ \mathbb{E}[X_i X_i']^{-1},$$

and so the asymptotic variance for the slope coefficient is

$$\Sigma_{11} = \frac{\mathrm{Var}[U_i]}{\mathrm{Var}[X_{i1}]} \equiv \frac{\text{"noise"}}{\text{"signal"}}.$$

Signal: more variation in the x-axis makes it easier to determine slope. **Noise:** more variation in the y-axis makes it hard to determine slope.

2.2.2 Asymptotic Variance of Subvectors

Theorem 2.4 (FWL for variance matrices)**.** *Write $X_i = [X_{i1}', X_{i2}']'$, $\beta \coloneqq [\beta_1', \beta_2']'$, and the asymptotic variance of B_1 as Σ_1. Then,*

$$\Sigma_1 = \mathbb{E}[V_i V_i']^{-1}\ \mathbb{E}[U_i^2 V_i V_i']\ \mathbb{E}[V_i V_i']^{-1},$$

where $V_i \coloneqq X_{i1} - \mathbb{L}[X_{i1}|X_{i2}]$.

[2] We usually start with the homoskedasticity case for intuition.

Proof. Note that $B_1 = (\hat{V}'\hat{V})^{-1}\hat{V}'Y$, where $\hat{V}_i = X_{i1} - CX_{i2}$ with $C \xrightarrow{p} \mathbb{E}[X_{i2}X_{i2}']^{-1}\,\mathbb{E}[X_{i2}X_{i1}']$. Thus,

$$\sqrt{n}(B_1 - \beta_1) = \left(\frac{1}{n}\sum \hat{V}_i\hat{V}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum \hat{V}_i U_i\right).$$

One can show

$$\frac{1}{\sqrt{n}}\sum \hat{V}_i U_i \longrightarrow 0,$$

and then the usual argument can be applied to conclude. □

2.2.3 Short and Long Regressions

$$\begin{aligned} Y_i &= \beta_{1s}X_{i1} + U_{is} \\ Y_i &= \beta_{1l}X_{i1} + X_{i2}'\beta_{2l} + U_{il}. \end{aligned}$$

How do the asymptotic variances of B_{1s} and B_{1l} compare? Let's assume X_{i1} is a scalar, so

$$\sigma_s^2 \equiv \frac{\mathbb{E}[U_{is}^2]}{\mathbb{E}[X_{i1}^2]}, \quad \sigma_l^2 \equiv \frac{\mathbb{E}[U_{il}^2]}{\mathbb{E}[V_i^2]}$$

where $V_i \equiv X_{i1} - \mathbb{L}[X_{i1}|X_{i2}]$ and we assume homoskedasticity.

Let's compare the two variances in two cases when the short and long coefficients are the same (so that we can compare apples to apples):

(i) $\beta_{2l} = 0$. This gives $\mathbb{E}[U_{il}^2] = \mathbb{E}[U_{is}^2]$. But since $\mathbb{E}[V_i^2] \le \mathbb{E}[X_{i1}^2]$, we have $\sigma_l^2 \ge \sigma_s^2$; X_{i2} reduces the independent variation in X_{i1} and hence increases variance.

(ii) X_{i1} and X_{i2} orthogonal (think controlling for pre-treatment covariates in a RCT). Then $\mathbb{E}[X_{i1}^2] = \mathbb{E}[V_i^2]$. But since $\mathbb{E}[U_{is}^2] = \mathbb{E}[U_{il}^2] + \mathbb{E}[(X_{i2}'\beta_{2l})^2]$, $\sigma_l^2 \le \sigma_s^2$; X_{i2} reduces residual variation in Y_i and hence reduces variance.

In the general case, it is a horse race between the two special cases: independent variation vs. residual variation.

2.3 Clustering

Definition 2.5. **Clustering** assumes independence across clusters ($c_i \neq c_j$) but arbitrary dependence within.

Example 2.6. Cluster at the individual level in panel data. Cluster based on geography in **cluster sampling**, e.g., first sample city, then sample individuals.

But all these decisions should be motivated by our *model*:

Example 2.7. If we think there are "common shocks" to wages in the industry, then we should cluster by industry.

Example 2.8. What is the source of uncertainty in administrative data that observe every person? The answer depends on the thought experiment. Think sampling states of the world (common in time series analysis) vs sampling people within a fixed state of the world.

Example 2.9 (Similarity vs dependence). One should expect individuals in the same city to have similar covariates. But this doesn't necessarily mean we should cluster by city. If the individuals are sampled independently (as in CPS, and not using a clustered sampling method), then there is of course no dependence in the sampling process.

2.4 The Structure of OLS with Clusters

Suppose we have observations $i = 1, \dots, n$ according to clusters $c_i \in \{1, \dots, m\}$. Then,

$$B - \beta = \left(\sum_{i=1}^{n} X_i X_i' \right)^{-1} \left(\sum_{i=1}^{n} X_i U_i \right) = \left(\frac{1}{m} \sum_{j=1}^{m} G_j \right)^{-1} \left(\frac{1}{m} \sum_{j=1}^{m} R_j \right),$$

with $G_j \coloneqq \sum_{i:c_i=j} X_i X_i'$ and $R_j \coloneqq \sum_{i:c_i=j} X_i U_i$. Now it's like we're working with new observations indexed by cluster j.

While G_j and R_j are not identically distributed across j (the cluster sizes can be different), a version of WLLN can still be applied to show that as $m \to \infty$,

$$\frac{1}{m} \sum_{j=1}^{m} G_j \xrightarrow{\text{p}} \Gamma$$

for some Γ, and similarly for $m^{-1} \sum_{j=1}^{m} R_j$. Consistency thus follows:

$$B - \beta = \left(\frac{1}{m} \sum_{j=1}^{m} G_j \right)^{-1} \left(\frac{1}{m} \sum_{j=1}^{m} R_j \right) \xrightarrow{\text{p}} 0.$$

Similarity, a version of CLT gives

$$\frac{1}{\sqrt{m}} \sum_{j=1}^{m} R_j \xrightarrow{\text{d}} \mathcal{N}(0, \Omega), \quad \Omega \coloneqq \lim_{\tilde{m} \to \infty} \frac{1}{\tilde{m}} \sum_{k=1}^{\tilde{m}} \mathbb{E}[R_k R_k'],$$

so

$$\sqrt{m}(B - \beta) \xrightarrow{\text{d}} \mathcal{N}(0, \underbrace{\Gamma^{-1} \Omega \Gamma^{-1}}_{=: \Sigma_{\text{CL}}}).$$

We may estimate Σ_{CL} with the **cluster-robust variance estimator** (CRVE):

$$V_{\text{CRVE}} \coloneqq \left(\frac{1}{m} \sum_{j=1}^{m} G_j \right)^{-1} \left(\frac{1}{m} \sum_{j=1}^{m} \hat{R}_j \hat{R}_j' \right) \left(\frac{1}{m} \sum_{j=1}^{m} G_j \right)^{-1}, \quad \hat{R}_j \coloneqq \sum_{i:c_i=j} X_i \hat{U}_i.$$

Remark 2.10. It is usually thought — and supported by theory under simplifying assumptions and Monte Carlo simulations — that the finite sample bias of CRVE is downward (CIs are too narrow). A common fix is to use the following finite-sample adjustment:

$$V_{\text{CRVE,ADJ}} \coloneqq \left(\frac{m}{m-1} \right) \left(\frac{n-1}{n-d_x} \right) V_{\text{CRVE}}.$$

While this certainly makes the standard errors larger, the justification seems a bit ad hoc.

3 Panel Data

We usually assume the panel is **balanced**, i.e., we have data on the same n units at each of d_t periods. This makes the theory much more intuitive.

Remark 3.1. We sometimes balance the panel before analysis, but note selective attrition can change the interpretation of results. ☕

3.1 Pooled Regressions

Given data $\{Y_{i1}, X_{i1}, \dots, Y_{id_t}, X_{id_t}\}_{i=1}^n$ on n units at each of d_t periods, and with X_{it} a d_x vector for each t, the **pooled OLS estimator** is an OLS of Y_{it} on X_{it} treating all (i, t) symmetrically:

$$B_{\text{PLS}} \coloneqq \left(\frac{1}{nd_t} \sum_{i,t} X_{it} X_{it}' \right)^{-1} \left(\frac{1}{nd_t} \sum_{i,t} X_{it} Y_{it} \right).$$

Remark 3.2 (An Asymmetry). We usually analyze short panels where d_t is fixed ("small"). Asymptotics come from the i dimension, as $n \to \infty$.

Similarly, note that $\mathbb{E}[X_{it}]$ is always taken over i, for fixed t. The grand average will be written as $\mathbb{E}[d_t^{-1} \sum_t X_{it}]$, and the time-average of a variable will be written as $\bar{X}_i \coloneqq d_t^{-1} \sum_t X_{it}$. ☕

The population analogue of the pooled OLS estimator is

$$\beta \coloneqq \arg\min_b \mathbb{E}\left[\frac{1}{d_t} \sum_t (Y_{it} - X_{it}' b)^2 \right] = \mathbb{E}\left[\frac{1}{d_t} \sum_t X_{it} X_{it}' \right]^{-1} \mathbb{E}\left[\frac{1}{d_t} \sum_t X_{it} Y_{it} \right],$$

with the corresponding frequentest thought experiment being sampling units i from the population and observing the entire t series. As before, the defining orthogonality condition for β, sometimes called **pooled orthogonality**, is

$$\mathbb{E}\left[\frac{1}{d_t} \sum_t X_{it} U_{it} \right] = 0.$$

We may stack the time dimension:

$$Y_i = \begin{bmatrix} Y_{i1} \\ \vdots \\ Y_{id_t} \end{bmatrix}, \quad X_i = \begin{bmatrix} X_{i1}' \\ \vdots \\ X_{id_t}' \end{bmatrix}, \quad U_i = \begin{bmatrix} U_{i1} \\ \vdots \\ U_{id_t} \end{bmatrix}$$

and write

$$Y_i = X_i \beta + U_i, \quad \mathbb{E}[X_i' U_i] = 0, \quad B_{\text{PLS}} = \left(\sum X_i' X_i \right)^{-1} \left(\sum X_i' Y_i \right).$$

This is a special case of what we did in clustering. We can conclude as before that

$$\sqrt{n}(B_{\text{PLS}} - \beta) = \left(\frac{1}{n} \sum X_i' X_i \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum X_i' U_i \right) \xrightarrow{\text{d}} \mathcal{N}(0, \Sigma_{\text{PLS}}),$$

where $\Sigma_{\text{PLS}} \coloneqq \mathbb{E}[X_i' X_i]^{-1} \mathbb{E}[X_i' U_i U_i' X_i] \mathbb{E}[X_i' X_i]^{-1}$. The cluster-robust variance estimator at the unit level is

$$V_{\text{CRVE}} = \left(\frac{1}{n} \sum X_i' X_i \right)^{-1} \left(\frac{1}{n} \sum X_i' \hat{U}_i \hat{U}_i' X_i \right) \left(\frac{1}{n} \sum X_i' X_i \right)^{-1}.$$

3.2 Unit Fixed Effects

Call the unit id in panel data A_i (Alias). This is a slightly unusual variable: in each realization, A_i takes exactly n categorical values. Conditioning on A_i conditions on all time-invariant aspects of unit i.

A linear *model* specification with unit fixed effects (UFE) is of the following form:

$$\mathbb{E}[Y_{it}|X_{i1},\dots,X_{id_t},A_i] = X_{it}'\beta + \alpha(A_i)$$

with unknown parameters β and α. A implication of this model specification is **strict orthogonality**: $\mathbb{E}[U_{it}X_{is}] = 0_{d_x}$ for all t,s. Note that this is stronger than pool orthogonality.

Instead of adding in n regressors for the nonlinear α, we apply a **within transformation** on $Y_{it} = X_{it}'\beta + \alpha(A_i) + U_{it}$ to de-mean *within* unit, over time:

$$\dot{Y}_{it} = \dot{X}_{it}'\beta + \dot{U}_{it}.$$

With the stacking notation introduced in the previous subsection, this is $\dot{Y}_i = \dot{X}_i\beta + \dot{U}_i$. By strict orthogonality, $\dot{X}_{it}$ and $\dot{U}_{it}$ are orthogonal:

$$\mathbb{E}[\dot{X}_{it}\dot{U}_{it}] = \mathbb{E}[X_{it}U_{it}] - \frac{1}{d_t}\sum_{s=1}\mathbb{E}[X_{is}U_{it}] - \frac{1}{d_t}\sum_{s=1}\mathbb{E}[X_{it}U_{is}] + \frac{1}{d_t^2}\sum_{r,s}\mathbb{E}[X_{is}U_{ir}] = 0.$$

It is then straightforward to see that $\dot{X}_{it}$ and $\dot{U}_{it}$ are also pooled orthogonal, giving the normal equations

$$\mathbb{E}[\dot{X}_i'\dot{Y}_i] = \mathbb{E}[\dot{X}_i'(\dot{X}_{it}'\beta + \dot{U}_i)] = \mathbb{E}[\dot{X}_i'\dot{X}_i]\beta.$$

Assuming the Gram matrix is invertible, we have

$$\beta = \mathbb{E}[\dot{X}_i'\dot{X}_i]^{-1}\,\mathbb{E}[\dot{X}_i'\dot{Y}_i] = \mathbb{E}\left[\frac{1}{d_t}\sum_s \dot{X}_{is}\dot{X}_{is}'\right]^{-1}\mathbb{E}\left[\frac{1}{d_t}\sum_s \dot{X}_{is}\dot{Y}_{is}\right].$$

Thus,

Proposition 3.3. *The UFE estimator (or within estimator) B_{UFE} is precisely the pooled OLS estimator of $\dot{Y}_{it}$ on $\dot{X}_{it}$.*

Remark 3.4. We can also remove the $\alpha(A_i)$ by defining say $\Delta y_{it} = y_{it} - y_{i,t-1}$, but the within transformation is often preferred due to better variance properties. Note that we may also derive the UFE estimator using FWL. ☕

Remark 3.5. We can estimate $\alpha(A_i)$ by regressing $Y_{it} - X_{it}'B_{\text{UFE}}$ on A_i. But since d_t is usually small, the estimates of $\alpha(A_i)$ are not very precise. ☕

3.2.1 Collinearity Concerns

We assumed above that $\mathbb{E}[\dot{X}_i'\dot{X}_i]$ is invertible. This is the case if and only if there is no $c \neq 0_{d_x}$ such that $\dot{X}_i c = 0_{d_t}$ almost surely. That is, no component of $\dot{X}_{it}$ can be a perfect linear combination of the others, for all t. In particular, X_{it} cannot include time-invariant variables since the corresponding column of $\dot{X}_i$ would be zero (these time-invariant variables are already controlled for by $\alpha(A_i)$). This does not mean that X_{it} needs to very for each i, just that it can't be constant for all i.

For the individuals with no variation in X_{itk}, the corresponding de-meaned regressors are zero, so they don't contribute to the estimation of β_k. The UFE estimate is determined solely by the "movers" whose X_{itk} varies over time.

3.2.2 Asymptotic Distribution and Variance of the UFE Estimator

We've shown that a UFE is precisely a pooled OLS estimator on the de-meaned data. Thus,

$$\sqrt{n}(B_{\text{UFE}} - \beta) = \left(\frac{1}{n}\sum \dot{X}_i'\dot{X}_i\right)^{-1}\left(\frac{1}{\sqrt{n}}\sum \dot{X}_i'\dot{U}_i\right) \xrightarrow{\text{d}} \mathcal{N}(0, \Sigma_{\text{UFE}}),$$

where $\Sigma_{\text{UFE}} = \mathbb{E}[\dot{X}_i'\dot{X}_i]^{-1}\,\mathbb{E}[\dot{X}_i'\dot{U}_i\dot{U}_i'\dot{X}_i]\,\mathbb{E}[\dot{X}_i'\dot{X}_i]^{-1}$ and we may estimate Σ_{UFE} using V_{CRVE} with within-transformed variables.

Example 3.6 (A Special Case). X_{it} scalar, U_{it} serially uncorrelated, homoskedastic with $\mathbb{E}[U_{it}^2] = \sigma^2$, then

$$\Sigma_{\text{UFE}} = \frac{\sigma^2}{\mathbb{E}\left[\sum_{t=1}^{d_t}\dot{X}_{it}^2\right]} = \frac{\text{noise (within variation)}}{\text{signal}}.$$

3.3 Two Way Fixed Effects

We may also add **time fixed effects** (TFE). By FWL this is equivalent to a pooled OLS on the **between**-transformed data, with the cross-sectional average at each time t subtracted off.

A **two-way fixed effects** (TWFE) model includes both UFE and TFE:

$$\mathbb{E}[Y_{it}|X_{i1},\dots,X_{id_t},A_i,T_t] = X_{it}'\beta + \gamma_t + \alpha(A_i).$$

This is often written as $Y_{it} = X_{it}'\beta + \gamma_t + \alpha_i + U_{it}$. We apply the same within transformation as before to get:

$$\dot{Y}_{it} = \dot{X}_{it}'\beta + \dot{\gamma}_t + \dot{U}_{it}', \qquad \dot{\gamma}_t \coloneqq \gamma_t - \frac{1}{d_t}\sum_{s=1}^{d_t}\gamma_s.$$

This can ten be rewritten as

$$\dot{Y}_i = \dot{X}_i\beta + \dot{\gamma} + \dot{U}_i = \begin{bmatrix}\dot{X}_i & \mathrm{Id}_{d_t}\end{bmatrix}\begin{bmatrix}\beta\\ \dot{\gamma}\end{bmatrix} + \dot{U}_i \eqqcolon W_i\delta + \dot{U}_i.$$

As before, our model assumption gives the orthogonality conditions $\mathbb{E}[\dot{X}_i'\dot{U}_i] = 0_{d_x}$ and $\mathbb{E}[\mathrm{Id}_{d_t}'\,\dot{U}_i] = \mathbb{E}[\dot{U}_i] = 0$. Thus we have $\mathbb{E}[W_i'\dot{U}_i] = 0$ and so

$$\delta = \mathbb{E}[W_i'W_i]^{-1}\,\mathbb{E}[W_i'\dot{Y}_i].$$

The corresponding estimator is the **TWFE estimator**.

Remark 3.7. To avoid collinearity, we now also need each component of X_{it} to vary across both units and time.

We may again use FWL to get an expression of the coefficients on X_{it} in terms of a regression of **doubly de-meaned** data: As before we use $\dot{Y}_{it}$ and $\dot{X}_{it}$ to denote the within-transformed data. We first regress $\dot{X}_{it}$ onto time indicators to get

$$\ddot{X}_{it} := \dot{X}_{it} - \mathbb{E}[\dot{X}_{it}] = X_{it} - \underbrace{\frac{1}{d_t}\sum_{s=1}^{d_t} X_{is}}_{\text{time mean}} - \underbrace{\mathbb{E}[X_{it}]}_{\text{unit mean}} + \underbrace{\mathbb{E}\left[\frac{1}{d_t}\sum_{s=1}^{d_t} X_{is}\right]}_{\text{grand mean}}.$$

Then, regressing Y_{it} (or equivalently $\dot{Y}_{it}$ or $\ddot{Y}_{it}$) onto $\ddot{X}_{it}$ gives the same coefficient as the TWFE estimator.

3.4 Mundlak Regressions

Proposition 3.8 (One way Mundlak)**.** *The following regressions give the same coefficient on X_{it}:*

(i) UFE of Y_{it} on X_{it}.

(ii) Pooled OLS of Y_{it} on X_{it} and $\bar{X}_i := \sum_{t=1}^{d_t} X_{it}/d_t$. This is called a ***one-way Mundlak*** *(OWM) regression.*

Proof. We apply FWL to the OWM regression: Write $X_{it} = \eta\bar{X}_i + V_{it}$. It is easy to see that $\eta = \mathrm{Id}_{d_x}$, so $V_{it} = X_{it} - \bar{X}_i = \dot{X}_{it}$.[3] Thus β_{OWM} is the coefficient from a pooled regression of Y_{it} onto $\dot{X}_{it}$, which is precisely the UFE estimator. □

Proposition 3.9 (Two way Mundlak)**.** *If the panel is balanced, then the following regressions give the same coefficients on X_{it}:*

(i) TWFE of Y_{it} on X_{it}.

(ii) Pooled OLS of Y_{it} on X_{it}, $\bar{X}_i$, $\mathbb{E}[X_{it}]$, and a constant.[4] *This is called a* ***two-way Mundlak*** *(TWM) regression.*

Proof. Effectively, the argument boils down to noting that the residuals of $\bar{X}_i$ and $\mathbb{E}[X_{it}]$ in a regression of a constant ($\bar{X}_{it} - \mathbb{E}[\bar{X}_i]$ and $\mathbb{E}[X_{it}] - \mathbb{E}[\bar{X}_i]$) are orthogonal. These are the residuals of regressing $\mathbb{E}[X_{it}]$ and $\bar{X}_i$ on a constant.

We first regress X_{it} onto $\bar{X}_i$, X_{it}, and a constant:

$$X_{it} = \eta_1\bar{X}_i + \eta_2\,\mathbb{E}[X_{it}] + \eta_3 + V_{it}.$$

Subtracting the pooled mean gives:[5]

$$X_{it} - \mathbb{E}[\bar{X}_i] = \eta_1(\bar{X}_i - \mathbb{E}[\bar{X}_i]) + \eta_2\underbrace{(\mathbb{E}[X_{it} - \mathbb{E}[\bar{X}_i]])}_{\mathbb{E}[\dot{X}_{it}]} + V_{it} - 0.$$

[3] One way to see this mechanically is to write out the formula for η and use TLE.

[4] A constant is not necessary for the one way Mundlak.

[5] Or think FWL within FWL.

Note that $\bar{X}_i - \mathbb{E}[\bar{X}_i]$ and $\mathbb{E}[\dot{X}_{it}]$ are pooled orthogonal:

$$\mathbb{E}\left[\frac{1}{d_t}\sum_s (\bar{X}_i - \mathbb{E}[\bar{X}_i])\,\mathbb{E}[\dot{X}_{is}]'\right] = \frac{1}{d_t}\sum_s (\mathbb{E}[\bar{X}_i] - \mathbb{E}[\bar{X}_i])\,\mathbb{E}[\dot{X}_{is}]' = 0.$$

Thus by the characterization of short/long regression, η_1 and η_2 can be solved for in isolation. They turn out to both be identity matrices, so $V_{it} = X_{it} - \bar{X}_i - \mathbb{E}[X_{it}] + \mathbb{E}[\bar{X}_i] = \ddot{X}_{it}$. □

We have the following special case of the two-way Mundlak regression:

Corollary 3.10. *Suppose $X_{itk} = W_{ik}Z_{tk}$ for variables W_{ik} and Z_{tk} that vary at the i and t level only. If the panel is balanced, then the following regressions give the same coefficient on X_{it}:*

(i) TWFE of Y_{it} on X_{it}.

(ii) Y_{it} onto a constant, W_i, Z_t, and X_{it}. This is a regression with ***two-way product*** *(TWP).*

Proof. Note that

$$\mathbb{E}[X_{it}] = \mathbb{E}[W_i]Z_t = (\text{constant}) \cdot Z_t, \quad \bar{X}_i = W_i\left(\frac{1}{d_t}\sum_t Z_t\right) = (\text{constant}) \cdot W_i.$$

Scaling W_i or Z_t by a constant does not change the coefficients on X_{it}. □

4 Single Events

Notation:

- Event happens at t^*, affecting periods $t \geq t^*$.
- Event affected units have $G_i = 1$, other units have $G_i = 0$.
- Observations $Y_{i1}, \dots, Y_{id_t}$. Write $Y_{it}(0)$ and $Y_{it}(1)$ for the potential outcomes at time t if treatment event had not / had happened at t^*.

We will focus on the ATT:

$$\text{ATT}_t := \mathbb{E}[Y_{it}(1) - Y_{it}(0) | G_i = 1]$$

Note that the first part $\mathbb{E}[Y_{it} | G_i = 1]$ is directly identified, whereas the second part $\mathbb{E}[Y_{it}(0) | G_i = 1]$ is the counterfactual.

4.1 The Two-Period Case

We have $d_t = t^* = 2$.

Assumption 4.1 (No Anticipation, NA). Pre-period outcomes for the treated group are as if no treatment had happened:

$$\mathbb{E}[Y_{it}(1) | G_i = 1] = \mathbb{E}[Y_{it}(0) | G_i = 1].$$

Assumption 4.2 (Common Trends, CT). Write $\Delta(g) := \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) | G_i = g]$. We assume untreated outcomes would have evolved in the same way in both groups: $\Delta(0) = \Delta(1)$.

Note that $\Delta(0) = \mathbb{E}[Y_{i2} - Y_{i1} | G_i = 0]$ is directly identified, and $\Delta(1) = \Delta(0)$ is identified by the common trends assumption. Thus by NA and CT,

$$\mathbb{E}[Y_{i2}(0) | G_i = 1] = \mathbb{E}[Y_{i1}(0) | G_i = 1] + \Delta(1) = \mathbb{E}[Y_{i1} | G_i = 1] + \Delta(0),$$

is also identified. Hence the ATT can be identified by a DID contrast:

$$\begin{aligned}
\text{ATT}_2 &:= \mathbb{E}[Y_{i2} | G_i = 1] - \mathbb{E}[Y_{i2}(0) | G_i = 1] \\
&= \mathbb{E}[Y_{i2} | G_i = 1] - \left(\mathbb{E}[Y_{i1} | G_i = 1] + \Delta(0)\right) \\
&= \mathbb{E}[Y_{i2} - Y_{i1} | G_i = 1] - \mathbb{E}[Y_{i2} - Y_{i1} | G_i = 0].
\end{aligned}$$

4.2 Multiple Time Periods

Assumption 4.3 (NA).

$$\mathbb{E}[Y_{it}(1) | G_i = 1] = \mathbb{E}[Y_{it}(0) | G_i = 1], \quad \forall t < t^*.$$

Assumption 4.4 (CT).

$$\Delta_{s \to t}(0) = \Delta_{s \to t}(1), \quad \forall s, t$$

where $\Delta_{s \to t}(g) := \mathbb{E}[Y_{it}(0) - Y_{is}(0) | G_i = g]$.

Remark 4.5. Assumption CT is effectively a random assignment assumption: $Y_{it}(0) - Y_{is}(0)$ is independent of G_i. It is worth thinking why one might accept this but reject the assumption that levels are independent of G_i.

4.2.1 Pre-Trends

The DID model is **over identified**:

$$\textsc{att}_t = \mathbb{E}[Y_{it} - Y_{is}|G_i = 1] - \mathbb{E}[Y_{it} - Y_{is}|G_i = 0] =: \textsc{did}_{s\to t}$$

for each s. Thus for $s_1, s_2, t < t^*$, NA and CT imply

$$\textsc{did}_{s_2\to s_1} = \textsc{did}_{s_2\to t} - \textsc{did}_{s_1\to t} = \Delta_{s_2\to s_1}(1) - \Delta_{s_2\to s_1}(0) = 0.$$

This is a testable implication of the model, and is often used as a **placebo test** for the validity of the common trends assumption.

Remark 4.6 (A bias-variance trade-off). Longer pre-periods means more comparisons can be made, both for testing CT and for improving precision by averaging over comparisons. With longer periods, however, Assumption CT is a stronger assumption that becomes less plausible. ☕

Remark 4.7 (A trade-off between NA and CT). NA becomes more plausible as we go further into the pre-period. However, CT becomes more objectionable as we go further back in time. ☕

4.3 Implementing DIDs with Linear Regression

Start with the saturated regression $\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma_t g$ and center around a base period, say $t_0 := t^* - 1$:

$$\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma_{t_0} g + \beta_t g, \quad \beta_t := \gamma_t - \gamma_{t_0}.$$

Then, the coefficients on the interactions are DID contrasts with $t_0 = t^* - 1$:

$$\begin{aligned}\textsc{did}_{t_0\to t} &:= \mathbb{E}[Y_{it} - Y_{it_0}|G_i = 1] - \mathbb{E}[Y_{it} - Y_{it_0}|G_i = 0]\\ &= [(\tau_t + \gamma_{t_0} + \beta_t) - (\tau_{t_0} + \gamma_{t_0})] - [\tau_t - \tau_{t_0}] = \beta_t.\end{aligned}$$

Under Assumptions NA and CT,

$$\beta_t = \textsc{did}_{t_0\to t} = \begin{cases}\textsc{att}_t, & t \geq t^*\\ 0, & t < t^*\end{cases}.$$

Remark 4.8. By the saturated regression logic, dropping pre-periods $t < t_0$ does not change the $\textsc{did}_{t_0\to t}$ estimates. ☕

Note that the interaction terms are $X_{it} := \left[G_i \mathbb{1}\{t = s\}\right]_{s\neq t_0}$, so this is a two-way product regression. Thus by the Mundlak logic, this is equivalent to a TWFE regression of Y_{it} onto X_{it} if the panel is balanced.

One can also arrive at a TWFE specification directly: Write $\mathbb{E}[Y_{it}(0)|A_i] = \alpha(A_i) + \tau_t$. This implies unit and thus group level common trends. Then, assuming in addition that $\mathbb{E}[Y_{it}(1) - Y_{it}(0)|A_i, G_i] = \mathbb{E}[Y_{it}(1) - Y_{it}(0)|G_i]$, we have

$$\mathbb{E}[Y_{it}|A_i, G_i] = \alpha(A_i) + \tau_t + \mathbb{E}[Y_{it}(1) - Y_{it}(0)|G_i]G_i.$$

So which to use? Controlling for unit effects in TWFE rely on the stronger assumption that *all* units have common trends, while group effects allow for some deviation within group. In the balanced case, both produce the same coefficient estimates. With a imbalanced panel, however, one might one to control for more factors that perhaps caused the imbalance, and one way to achieve this is by using unit FEs instead of group FEs. Thus TWFE is more widely used in practice.

4.3.1 Aggregating Pre-Periods

We can use more pre-periods contrasts to reduce variance. The corresponding linear regression implementation is same regression with pre-period group-time interactions dropped. The resulting coefficients on post-period interactions will be the averages of the corresponding DID contrasts with different pre-periods:

Proposition 4.9. *Suppose the panel is balanced. In a regression of Y_{it} on 1, G_i, $\left[\mathbb{1}\{t = s\}\right]_{s=1,\dots,d_t}$, and $\left[G_i\mathbb{1}\{t = s\}\right]_{s\geq t^*}$, the coefficients on the interaction terms are the averages of the corresponding DID contrasts with different pre-periods.*

Proof. Note that the following regressions are equivalent:

(i) Y_{it} on $1, G_i, X_{it} \coloneqq \left[\mathbb{1}\{t = 1\}, \dots, \mathbb{1}\{t = d_t\}, G_i\mathbb{1}\{t = t^*\}, \dots, G_i\mathbb{1}\{t = d_t\}\right]'$.

(ii) Y_{it} with UFE on $X_{it} \coloneqq \left[\mathbb{1}\{t = 1\}, \dots, \mathbb{1}\{t = d_t\}, G_i\mathbb{1}\{t = t^*\}, \dots, G_i\mathbb{1}\{t = d_t\}\right]'$, by the one way Mundlak regression characterization, noting that $\bar{X}_i = d_t^{-1}[1', G_i']'$.

(iii) Y_{it} TWFE on $X_{it} = \left[G_i\mathbb{1}\{t = t^*\}, \dots, G_i\mathbb{1}\{t = d_t\}\right]'$ (dropping time indicators).

(iv) Y_{it} pooled on 1, G_i, $\left[\mathbb{1}\{t = s\}\right]_{s\geq t^*}$, and $X_{it} \coloneqq \left[G_i\mathbb{1}\{t = t^*\}, \dots, G_i\mathbb{1}\{t = d_t\}\right]'$, by the characterization of the two-way product regression.

The last regression is saturated. The coefficients on $G_i\mathbb{1}\{t = t^*\}, \dots, G_i\mathbb{1}\{t = d_t\}$ are precisely the corresponding DID contrasts with the average of all pre-periods. □

The same logic shows that replacing post-period interactions with a single indicator $X_{it}\mathbb{1}\{t \geq t^*\}$ gives the average of all DID contrasts (equivalently, the DID contrast with all pre- and post-periods collapsed):

Proposition 4.10. *Suppose the panel is balanced. In a regression of Y_{it} on 1, G_i, $\left[\mathbb{1}\{t = s\}\right]_{s=1,\dots,d_t}$, and $G_i\mathbb{1}\{t \geq t^*\}$, the coefficient on the interaction term is the average of all DID contrasts.*

Proof. The proposed regression is a TWFE regression of Y_{it} on $X_{it} \coloneqq G_i\mathbb{1}\{t \geq t^*\}$, which is by the two-way product characterization equivalent to a regression of Y_{it} on 1, G_i, and $X_{it} \coloneqq G_i\mathbb{1}\{t \geq t^*\}$. This is a saturated regression, and the coefficient on interaction is the DID contrast on the average of all pre-periods and all post-periods,

$$\mathbb{E}\left[\frac{1}{t_1}\sum_{t=t^*}^{d_t} Y_{it} - \frac{1}{t_0}\sum_{t=0}^{d_t-1} Y_{it}\middle| G_i = 1\right] - \mathbb{E}\left[\frac{1}{t_1}\sum_{t=t^*}^{d_t} Y_{it} - \frac{1}{t_0}\sum_{t=0}^{d_t-1} Y_{it}\middle| G_i = 0\right].$$

□

4.4 Triple Differences

One might think that TennCare's effect is concentrated on those without children (who are not eligible for Medicaid). Thus there should be an *differential* change in employment between adults with and without children in Tennessee. This is captured by a **triple differences** contrast.

One way to implement triple differences is to simply change the outcome from Y_{it} to $\tilde{Y}_{it} := Y_{it}^1 - Y_{it}^0$, where Y_{it}^h denote the observed outcome for the subgroup $h \in \{0, 1\}$.

It is of course sufficient to assume CT for each of the two subgroups. The point of triple differences is that this can be relaxed somewhat:

Assumption 4.11 (CT).

$$\Delta_{s\to t}^h(g) := \mathbb{E}[Y_{it}^h(0) - Y_{is}^h(0)|G_i = g] = \delta_{1st}g + \delta_{2st}^h, \quad \forall g, h, s, t.$$

The term δ_{1st} captures the violation of common trends between $g = 0$ and $g = 1$. Note that we assume this violation is the same across subgroups $h \in \{0, 1\}$, and so we have common trends in the differenced outcome $\tilde{Y}_{it}(0) := Y_{it}^1(0) - Y_{it}^0(0)$:

$$\mathbb{E}[\tilde{Y}_{it}(0) - \tilde{Y}_{is}(0)|G_i = g] = \Delta_{s\to t}^1(g) - \Delta_{s\to t}^0(g) = \delta_{2st}^1 - \delta_{2st}^0$$

does not depend on g.

By exactly the same logic as before,

$$\widetilde{\textsc{did}}_{s\to t} = \textsc{did}_{s\to t}^1 - \textsc{did}_{s\to t}^0 = \begin{cases} 0, & t < t^* \\ \widetilde{\textsc{att}}_t, & t \geq t^* \end{cases},$$

where $\widetilde{\textsc{att}}_t = \textsc{att}_t^1 - \textsc{att}_t^0$ is the difference in treatment effects between the two subgroups. Note that $\widetilde{\textsc{att}}_t = \textsc{att}_t^1$ if we assume the treatment has no effect on subgroup $h = 0$.

Remark 4.12. Triple differences is often used as a placebo test for the common trends assumption, assuming the $\textsc{att}_t^0 = 0$. ☕

4.5 Conditional Common Trends

Assumption 4.13 (Conditional CT).

$$\mathbb{E}[Y_{it}(0) - Y_{is}(0)|G_i = g, W_i] = \mathbb{E}[Y_{it}(0) - Y_{is}(0)|G_i = 0, W_i].$$

Remark 4.14. This assumption is made usually with time-invariant covariates W_i. If W_i is time-varying, one might worry that this variation is caused by the treatment itself or causes the treatment. ☕

One can separately compute DID estimates for each $W_i = w$ group and then aggregate somehow. This has no bias but potentially large variance. Instead, we seek a linear regression implementation. It is common to specify a TWFE model, since covariates necessarily vary at a level finer than G_i:

$$\mathbb{E}[Y_{it}(0)|A_i, G_i, W_i] = \alpha(A_i) + W_i'\tau_t + \gamma_t G_i.$$

One component of W_i would typically be a constant, so that standard time fixed effects are still included as part of $W_i'\tau_t$. The conditional CT assumption imposes $\gamma_t = 0$.

Now, assuming treatment effects does not depend on W_i or A_i, we have

$$\mathbb{E}[Y_{it}|A_i, G_i, W_i] = \alpha(A_i) + W_i'\tau_t + \gamma_t G_i + \beta_t G_i \mathbb{1}[t \geq t^*],$$

where $\beta_t = \text{ATT}_t$.

4.6 Statistical Issues

In settings such as the TennCare example, it is common to cluster at the state level, since one might worry about state level shocks. But with only $n = 18$ clusters and a single treated state (so $\mathbb{P}[G_i = 1] = n^{-1} \to 0$), how should one think about inference? What is the right frequentist thought experiment?

In the TennCare paper, the authors reported bootstrapped SEs with a two-step cluster bootstrap (resample states, then resample CPS respondents within each year). Bootstrapping to some extent addresses the CPS sampling uncertainty, but what is the relevant uncertainty from the state level sampling?

Analyses of staggered events mitigate these issues somewhat.

4.7 Summary of Regression Specifications

(i) Y_{it} on 1, G_i, time FEs, and interactions centered at t_0, $\left[G_i \mathbb{1}\{t = s\}\right]_{s \neq t_0}$:

$$\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma g + \beta_t g.$$

Coefficients on interactions are $\text{DID}_{t_0 \to t}$.
In the balanced panel case, this is equivalent to a TWFE regression of Y_{it} on $\left[G_i \mathbb{1}\{t = s\}\right]_{s \neq t_0}$.

(ii) Assume $\mathbb{E}[Y_{it}(1) - Y_{it}(0)|A_i, G_i] = \mathbb{E}[Y_{it}(1) - Y_{it}(0)|G_i]$. Then,

$$\mathbb{E}[Y_{it}|A_i, G_i] = \alpha(A_i) + \tau_t + \mathbb{E}[Y_{it}(1) - Y_{it}(0)|G_i]G_i.$$

Thus a TWFE regression of Y_{it} on $\left[G_i \mathbb{1}\{t = s\}\right]_{s \neq t_0}$ gives $\text{DID}_{t_0 \to t}$ on the interactions.

(iii) Assuming conditional CT $\mathbb{E}[Y_{it}(0) - Y_{it}(1)|G_i = g, W_i] = \mathbb{E}[Y_{it}(1) - Y_{it}(0)|G_i = 0, W_i]$ and a TWFE model $\mathbb{E}[Y_{it}(0)|A_i, G_i, W_i] = \alpha(A_i) + W_i'\tau_t + \gamma_t G_i$, we have

$$\mathbb{E}[Y_{it}|A_i, G_i, W_i] = \alpha(A_i) + W_i'\tau_t + \gamma_t G_i + \beta_t G_i \mathbb{1}[t \geq t^*],$$

where $\beta_t = \text{ATT}_t$.

4.7.1 Summarizing DIDs

Assuming a balanced panel, we can summarize DID contrasts using the following regressions:

(i) Averaging pre-period contrasts for each fixed post-period by omitting pre-period interactions:
  - Y_{it} on 1, G_i, time FEs, and post-period interactions $\left[G_i \mathbb{1}\{t = s\}\right]_{s \geq t^*}$,
  - Y_{it} TWFE on post-period interactions $\left[G \mathbb{1}\{t = s\}\right]_{s \geq t^*}$.
  - Y_{it} pooled on 1, G_i, post-period time FEs $\left[\mathbb{1}\{t = s\}\right]_{s \geq t^*}$, and post-period interactions $\left[G_i \mathbb{1}\{t = s\}\right]_{s \geq t^*}$.

(ii) Averaging all DID contrasts (equivalently, collapsing into two periods):
  - Y_{it} on 1, G_i, time FEs, and $G_i \mathbb{1}\{t \geq t^*\}$.
  - Y_{it} TWFE on $G_i \mathbb{1}\{t \geq t^*\}$.
  - Y_{it} pooled on 1, G_i, and $G_i \mathbb{1}\{t \geq t^*\}$.

5 Staggered Events

Notation:

- Time $t = 1, \dots, d_t$.
- Groups (**cohorts**) $G_i \in \{1, 2, \dots, d_t, \infty\}$, the time that an event happens. $G_i = \infty$ for the never-treated.
- Post-treatment period is $t \geq G_i$.
- Write $Y_{it}(g)$ as the outcome that would have happened if the event date had been g.

The target parameter is the ATT of having event in period g relative to never having the event:

$$\textsc{att}_t(g) \coloneqq \mathbb{E}[Y_{it}(g) - Y_{it}(\infty)|G_i = g], \quad t \geq g.$$

As before, the problem is that $\mathbb{E}[Y_{it}(\infty)|G_i = g]$ is not observed.

Assumption 5.1 (NA).

$$\mathbb{E}[Y_{it}(g)|G_i = g] = \mathbb{E}[Y_{it}(\infty)|G_i = g], \quad \forall t < g, \forall g.$$

Assumption 5.2 (CT).

$$\Delta_{s \to t}(g) \coloneqq \mathbb{E}[Y_{it}(\infty) - Y_{is}(\infty)|G_i = g]$$

does not depend on g for all s, t.

We can use the same identification logic as before. For any $h > t$ (so that group h is untreated at time t) and any pre-period $s < g$, we have

$$\begin{aligned}
\mathbb{E}[Y_{it}(\infty)|G_i = g] &= \mathbb{E}[Y_{is}(\infty)|G_i = g] + \mathbb{E}[Y_{it}(\infty) - Y_{is}(\infty)|G_i = g] \\
&= \mathbb{E}[Y_{is}(\infty)|G_i = g] + \mathbb{E}[Y_{it}(\infty) - Y_{is}(\infty)|G_i = h] \\
&= \mathbb{E}[Y_{is}(\infty)|G_i = g] + \mathbb{E}[Y_{it}(h) - Y_{is}(h)|G_i = h] \\
&= \mathbb{E}[Y_{is}|G_i = g] + \mathbb{E}[Y_{it} - Y_{is}|G_i = h].
\end{aligned}$$

Thus,

$$\textsc{att}_t(g) = \textsc{did}_{s \to t}(h \to g), \quad \forall s < g, \forall h > t.$$

Remark 5.3. Similar tradeoff logic: NA more likely for smaller s, but CT less likely. Usually think $h = g + 1$ is better comparison than $h = \infty$, but may want to use $h \geq t + 2$ if worried about one-period anticipation. Perhaps more importantly, using $h = g + 1$ identifies only the treatment effect $\textsc{att}_t(g)$ for the first post period $t = g$. ☕

For the purpose of interpretation, it is often more sensible to use **relative time** $\textsc{attr}_r(g) \coloneqq \textsc{att}_{g+r}(g)$, especially for comparing $\textsc{attr}_t(g)$ across g. The relative time of an observation is $R_{it} \coloneqq t - G_i$.

Another complication arises in summarizing $\textsc{attr}_r(g)$ for each r: we can identify $\textsc{attr}_0(g)$ for each g, but $\textsc{attr}_{d_t-2}(g)$ only for $g = 2$. The *composition* of identifiable

$\text{ATTR}_r(g)$'s change over the relative time r. One solution is to add in further conditioning to ensure composition stays fixed:

$$\overline{\text{ATTR}}_{r|r_1,r_2} \coloneqq \sum_{g=2}^{d_t} \text{ATTR}_r(g)\, \mathbb{P}\left[G_i = g | \underbrace{G_i + r_1, G_i + r_2 \in \{1, \dots, d_t\}}_{\text{observed at relative times } r_1, \dots, r_2}\right].$$

This is often done implicitly through sample selection rules, by say having the panel run 10 years before the first event and 10 years after the last event, so that we can identify $\text{ATTR}_r(g)$ for $r = -10, \dots, 10$ for all g.

Depending on application, one may alternatively summarize the average effect over some treatment periods, say

$$\overline{\text{ATTR}}_{0:r}(g) \coloneqq \frac{1}{r+1} \sum_{s=0}^{r} \text{ATTR}_s(g).$$

A recipe for staggered event analysis:

(i) Pick target parameters, including placebos.
(ii) Find all good (h, s) comparisons for each (g, t).
(iii) Estimate $\text{ATT}_t(g)$ via averaged DID contrasts.
(iv) Combine into summary target parameter, converting to relative time.
(v) Bootstrap for inference.

Some reasons for preferring a linear regression implementation: bootstrapping is slow and annoying, and the weighting of different DID constrains may not be efficient for the asymptotic variance of the summary parameter of interest.

5.1 Linear Regression Implementation

For some fixed base cohort g_0, write

$$\mathbb{E}[Y_{it}|G_i = g] = \mathbb{E}[Y_{it}|G_i = g_0] + (\mathbb{E}[Y_{it}|G_i = g] - \mathbb{E}[Y_{it}|G_i = g_0]) \eqqcolon \tau_t + \theta_{tg}.$$

We again center around a base period t_0:

$$\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \theta_{t_0 g} + \beta_{tg}, \qquad \beta_{tg} \coloneqq \theta_{tg} - \theta_{t_0 g}.$$

Read the first two terms as the time and cohort fixed effects, and the last term as the cohort-time interaction. Then, the cohort-time interactions are the DID constants with the base time-cohort:

$$\begin{aligned} \text{DID}_{t_0 \to t}(g_0 \to g) &= \mathbb{E}[Y_{it} - Y_{it_0}|G_i = g] - \mathbb{E}[Y_{it} - Y_{it_0}|G_i = g_0] \\ &= \left(\tau_t + \theta_{t_0 g} + \beta_{tg} - \tau_{t_0} - \theta_{t_0 g}\right) - \left(\tau_t - \tau_{t_0}\right) = \beta_{tg}. \end{aligned}$$

The comparison with base cohort for any two time periods s, t can then be written as

$$\text{DID}_{s \to t}(g_0 \to g) = \text{DID}_{t_0 \to t}(g_0 \to g) - \text{DID}_{t_0 \to s}(g_0 \to g) = \beta_{tg} - \beta_{sg},$$

and the comparison between any two groups and times is

$$\begin{aligned}\textsc{did}_{s\to t}(h \to g) &= \textsc{did}_{t_0\to t}(h \to g) - \textsc{did}_{t_0\to s}(h \to g)\\ &= (\beta_{tg} - \beta_{th}) - (\beta_{sg} - \beta_{sh}).\end{aligned}$$

For the sake of interpretation, let's switch to relative time, $\tilde{\beta}_{(t-g)g} \coloneqq \beta_{tg}$, and center at relative time -1:

$$\begin{aligned}\mathbb{E}[Y_{it}|G_i = g] &= \tau_t + \theta_{t_0 g} + \tilde{\beta}_{(t-g)g}\\ &= \tau_t + (\theta_{t_0 g} + \tilde{\beta}_{-1g}) + \left(\tilde{\beta}_{(t-g)g} - \tilde{\beta}_{-1g}\right) = \tau_t + \gamma_g + \rho_{(t-g)g}.\end{aligned}$$

Now, relative time coefficients are DID contrasts with g_0 and relative time -1:

$$\begin{aligned}\textsc{did}_{(g-1)\to t}(g_0 \to g) &= \mathbb{E}[Y_{it} - Y_{i(g-1)}|G_i = g] - \mathbb{E}[Y_{it} - Y_{i(g-1)}|G_i = g_0]\\ &= (\tau_t + \gamma_g + \rho_{(t-g)g} - \tau_{(g-1)} - \gamma_g - \rho_{-1g}) - (\tau_t - \tau_{g-1}) = \rho_{(t-g)g},\end{aligned}$$

where $\rho_{-1g} = \tilde{\beta}_{-1g} - \tilde{\beta}_{-1g} = 0$ by construction. Thus, under Assumptions NA and CT,

$$\rho_{(t-g)g} = \textsc{did}_{(g-1)\to t}(g_0 \to g) = \begin{cases} 0, & g_0 > t, t < g \\ \textsc{att}_t(g) = \textsc{attr}_{t-g}(g), & g_0 > t, t \geq g \\ \text{junk}, & g_0 \leq t \end{cases}.$$

To avoid junk comparisons, we can set $g_0 = \infty$ or $g_0 = d_t$. And as before, the assumptions can be tested by looking at pre-period contrasts $\rho_{(t-g)g}$, $t - g < 0$.

5.2 Linear Regression Pitfalls

5.2.1 Collapsing Cohort-Relative Time Interactions

It is tempting to use uninteracted relative time indicators, with $\rho_{(t-g)}$ same for all g, by specifying a regression of Y_{it} on calendar time effects, cohort effects, and positive relative time indicators. But bad comparisons gets used implicitly in this specification.

To justify the uninteracted regression, we need some additional assumption, such as the following:

Assumption 5.4 (Homogeneous cohort treatment effects (HC)).

$$\textsc{attr}_r(g) = \textsc{attr}_r, \quad \forall g, r.$$

Under Assumption HC, $\rho_{(t-g)g}$ does not depend on g, and under NA and CT, is the coefficients on the positive relative time indicators.

For some intuition, consider adding a constant c to all observations made in say relative time r. Then, denoting the residual of the regression of the positive relative time indicator $\mathbb{1}\{t - G_i = r\}$ onto calendar time and cohort effects by V_{it}, we have

$$\mathbb{E}\left[\sum_t V_{it}(Y_{it} + \mathbb{1}\{t - G_i = r\}\, c)\right] = \mathbb{E}\left[\sum_t V_{it} Y_{it}\right] + c\, \underbrace{\mathbb{E}\left[\sum_t \mathbb{1}\{t - G_i = r\}\, V_{it}\right]}_{=0 \text{ by orthogonality}}.$$

Thus this operation does not change the estimates for say $\textsc{attr}_2$ — and it should not, since any observation made at relative time $r = 0$ gives a junk DID contrast.

However, if we were to only add the constant on observations made in a relative time r for a particular cohort g, then the second term will not vanish. This shows the problem with the uninteracted regression when HC is violated. (With positive relative time and group interactions, it will. This is to be expected, since NA and CT are respected, and any observation at relative time $r = 0$ gives a junk contrast that should not be used.)

5.2.2 Dynamic Specifications with Uninteracted Relative Time

An even more common specification is to include time effects, cohort effects, and all positive and negative relative time effects. This will not improve the situation; the coefficients on positive times are still contaminated, and the coefficients on negative times are hard to interpret. Some interpretation can presumably be done with Assumptions NA/CT/HC, but this sort of defeats the purpose — presumably including negative relative time indicators is to test NA and CT.

5.2.3 Collinearity issues

It is not sufficient to just leave out relative time -1.

If there is a never-treated cohort $G_i = +\infty$, then cohort effect is perfectly collinear with the relative time $-\infty$ effect. This is implicitly dropped, but then effects become relative to the combination of $-\infty$ and -1. A solution is to group together a couple of periods (say the last two periods).

Also, we run into the age-period-cohort issue (which is clearly problematic in levels, but unfixed by turning to fixed effects).

5.2.4 Collapsing Relative Time

Another common practice is to collapse positive relative time into bins, with the motivation of reducing the number of parameters and improving precision. This is done by specifying a regression of Y_{it} on calendar effects, cohort/unit effects, and bins of relative time indicators. This, however, will also implicitly use bad DID contrast. One way to restore interpretability is to assume the following:

Assumption 5.5 (Static Treatment Effects (ST)). Let $\mathcal{R}_1, \dots, \mathcal{R}_k$ denote a partition of the positive relative times. Then,

$$\textsc{attr}_r(g) = \textsc{attr}_{r'}(g) =: \textsc{attr}_{\mathcal{R}_j}(g), \quad \forall r, r' \in \mathcal{R}_j, \forall j = 1, \dots, k.$$

Given NA/CT/HC, assumption ST implies that $\rho_{t-g} = \sum_j \mathbb{1}\{t - g \in \mathcal{R}_j\}\,\textsc{attr}_{\mathcal{R}_j}$. Thus jointly, these assumptions justify the regression above.

Do we believe these assumptions? One way to decide is to evaluate the consequences of these assumptions. Note that under NA/CT/HC/ST, we have

$$\mathbb{E}[Y_{it} - Y_{i(t-1)}|G_i = h] = \mathbb{E}[Y_{it}(\infty) - Y_{i(t-1)}(\infty)|G_i = h], \quad \forall h \geq t - 1.$$

Thus, the treatment effect can be estimated using just post-period contrasts:

$$\text{DID}_{(t-1)\to t}(h \to t) = \mathbb{E}[Y_{it} - Y_{i(t-1)}|G_i = h] - \underbrace{\mathbb{E}[Y_{it} - Y_{i(t-1)}|G_i = t]}_{=\mathbb{E}[Y_{it}(\infty)-Y_{i(t-1)}(\infty)|G_i=t]} = \text{ATT}_t(t).$$

This might seem implausible.

5.3 Imputation

The idea is to run two separate regressions. The first estimates the counterfactuals outcomes, and the second summarizes treatment effects.

Recall our saturated relative time specification:

$$\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma_g + \rho_{(t-g)g}.$$

Under Assumptions NA and CT, if we only use observations in the pre-periods (i.e., those with $R_{it} := t - G_i < 0$), then all of the interacted relative time coefficients are zero. Thus for these pre-period observations,

$$\mathbb{E}[Y_{it}(\infty)|G_i = g] = \mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma_g.$$

We can therefore estimate the counterfactual outcomes by simply regressing Y_{it} onto time and cohort effects using only pre-period observations. Note that we need observations of $G_i = \infty$ to estimate τ_{d_t}.

Next, define $\tilde{Y}_{it} := Y_{it} - \mathbb{E}[Y_{it}(\infty)|G_i] = Y_{it} - \tau_t - \gamma_g$ and note that

$$\mathbb{E}[\tilde{Y}_{it}|G_i = g] = \begin{cases} 0, & t < g \\ \text{ATT}_t(g), & t \geq g \end{cases}.$$

We can then regress $\tilde{Y}_{it}$ onto a saturated set of relative time indicators $\{\mathbb{1}\{R_{it} \in \mathcal{R}_j\}\}_j$ to get estimates

$$\begin{aligned} \frac{\sum_t \mathbb{E}[\tilde{Y}_{it}\mathbb{1}\{R_{it} \in \mathcal{R}_j\}]}{\sum_t \mathbb{E}[\mathbb{1}\{R_{it} \in \mathcal{R}_j\}]} &= \frac{\sum_t \mathbb{E}[\mathbb{E}[\tilde{Y}_{it}|G_i]\mathbb{1}\{R_{it} \in \mathcal{R}_j\}]}{\sum_t \mathbb{E}[\mathbb{1}\{R_{it} \in \mathcal{R}_j\}]} \\ &= \frac{\sum_t \mathbb{E}[\text{ATT}_t(G_i)\mathbb{1}\{t \geq G_i\}\,\mathbb{1}\{R_{it} \in \mathcal{R}_j\}]}{\sum_t \mathbb{E}[\mathbb{1}\{R_{it} \in \mathcal{R}_j\}]}. \end{aligned}$$

So regressing $\tilde{Y}_{it}$ onto positive relative time indicators give weighted averages of various $\text{ATT}_t(g)$'s, regardless of whether these relative time indicators are interacted with cohort indicators or grouped into binds of relative times.

5.4 Staggered DID Summary

Clean contrast:

$$\text{ATT}_t(g) = \text{DID}_{s\to t}(h \to g), \quad \forall s < g \leq t < h.$$

Regression implementation:

$$\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma_g + \sum_{h\neq g_0}\sum_{s\neq t_0} \beta_{sh}\mathbb{1}\{t = s\}\,\mathbb{1}\{g = h\}$$
$$\mathbb{E}[Y_{it}|G_i = g] = \tau_t + \gamma_g + \sum_{r\neq -1}\sum_{h\neq g_0} \rho_{rh}\mathbb{1}\{t - g = r\}\,\mathbb{1}\{g = h\}$$

$$\rho_{(t-g)g} = \textsc{did}_{(g-1)\to t}(g_0 \to g) = \begin{cases} 0, & g_0 > t, t < g \\ \textsc{att}_t(g) = \textsc{attr}_{t-g}(g), & g_0 > t, t \geq g \\ \text{junk}, & g_0 \leq t. \end{cases}.$$

Imputation:

(i) $\mathbb{E}[Y_{it}(\infty)|G_i = g]$ can be estimated using the following regression:

$$\mathbb{E}[Y_{it}|G_i = g, t < g] = \tau_t + \gamma_g.$$

(ii) Then define $\tilde{Y}_{it} := Y_{it} - \tau_t - \gamma_g$ and observe that

$$\mathbb{E}[\tilde{Y}_{it}|G_i = g] = \begin{cases} 0, & t < g \\ \textsc{att}_t(g), & t \geq g \end{cases},$$

so we can then regress $\tilde{Y}_{it}$ onto a saturated set of positive relative time indicators to get estimates of weighted averages of various $\textsc{att}_t(g)$'s.

6 Asymptotic Theory

6.1 Convergence in Probability

Definition 6.1. A sequence of random vectors B_n converges in probability to B if for all $\varepsilon > 0$,

$$\mathbb{P}\left[\|B_n - B\| > \varepsilon\right] \longrightarrow 0.$$

In this case, we write $B_n \xrightarrow{\text{p}} B$. And if B_n is an estimator of β and $B_n \xrightarrow{\text{p}} \beta$, then we say B_n is **consistent** for β.

Proposition 6.2. *$B_n \xrightarrow{\text{p}} B$ if and only if every component $B_{nj} \xrightarrow{\text{p}} B_j$.*

Theorem 6.3 (WLLN)**.** *Let $W_i = [W_{i1}, \dots, W_{id_b}]$ be iid vectors with $\mu \coloneqq \mathbb{E}[W_i]$ and $\sigma_k^2 \coloneqq \mathrm{Var}[W_{ik}] < \infty$ for each k. Then $B_n \coloneqq n^{-1} \sum_i W_i \xrightarrow{\text{p}} \mu$.*

Proof. Note that

$$\mathbb{E}[\|B_n - \mu\|^2] = \sum_{k=1}^{d_b} \mathbb{E}[(B_{nk} - \mu_k)^2] = \sum_k \mathrm{Var}[B_{nk}],$$

and

$$\mathrm{Var}[B_{nk}] = \mathrm{Var}\left[\frac{1}{n}\sum_i W_{ik}\right] = \frac{1}{n^2}\sum_i \mathrm{Var}[W_{ik}] = \frac{\sigma_k^2}{n}.$$

Thus,

$$\mathbb{P}\left[\|B_n - \mu\| > \varepsilon\right] = \mathbb{P}\left[\|B_n - \mu\|^2 > \varepsilon^2\right] < \frac{\mathbb{E}[\|B_n - \mu\|^2]}{\varepsilon^2} = \frac{1}{\varepsilon^2}\frac{1}{n}\sum_k \sigma_k^2 \longrightarrow 0.$$

□

In the last line we used Markov's inequality:

Proposition 6.4. *For any non-negative random variable X and any $a > 0$,*

$$\mathbb{P}[X > a] < \frac{\mathbb{E}[X]}{a}.$$

Remark 6.5 (Connection to Monte Carlo Simulation). Let $B_{nj} = f(W_{nj})$ where W_{nj} are iid. Then for each b,

$$\frac{1}{m}\sum_j \mathbb{1}\{B_{nj} \le b\} \xrightarrow{\text{p}} \mathbb{P}[f(W) \le b]$$

as $m \to \infty$ for any fixed n.

Example 6.6 (Convergence in probability does not imply convergence in moments). Consider B_n such that $B_n = n$ with probability $1/n$ and $B_n = 0$ otherwise. Then $B_n \xrightarrow{\text{p}} 0$, but $\mathbb{E}[B_n] = 1$ for all n.

However this implication can be restored if we assume B_n is bounded.

Example 6.7 (Convergence in moments does not imply convergence in probability). Let B be non-degenerate and let $B_n \coloneqq (-1)^n B$.

Definition 6.8. We say B_n **converges** to B **in** r**th mean** if $\mathbb{E}[|B_n - B|^r] \to 0$.

Note that Markov's inequality gives

$$\mathbb{P}[|B_n - B| > \varepsilon] < \frac{\mathbb{E}[|B_n - B|^r]}{\varepsilon^r} \longrightarrow 0.$$

Thus,

Proposition 6.9. *Convergence in rth mean ($r > 0$) implies convergence in probability.*

Again the converse does not hold; consider the counterexample with $B_n \in \{0, n\}$.

Remark 6.10. Recall the MSE decomposition, from which we know that to show consistency, it suffices to show that the estimator is asymptotically unbiased and has variance going to zero.

Theorem 6.11 (Continuous Mapping Theorem)**.** *Suppose $B_n \xrightarrow{\text{p}} B$ and $\mathbb{P}[B \in \mathcal{B}] = 1$. Let f be a vector-valued function that is continuous on $\mathcal{B}$. Then, $f(B_n) \xrightarrow{\text{p}} f(B)$.*

Proof. Fix $\varepsilon > 0$ and define

$$\mathcal{B}_\delta \coloneqq \{b : \text{ there exists an } a \text{ with } \|b - a\| < \delta \text{ but } \|f(b) - f(a)\| > \varepsilon\}.$$

This is the set on which f is close to discontinuous. Now,

$$\begin{aligned}\mathbb{P}\left[\|f(B_n) - f(B)\| > \varepsilon\right] &= \mathbb{P}\left[\|f(B_n) - f(B)\| > \varepsilon, \|B_n - B\| < \delta\right] \\ &\quad + \mathbb{P}\left[\|f(B_n) - f(B)\| > \varepsilon, \|B_n - B\| \geq \delta\right] \\ &\leq \mathbb{P}[B \in \mathcal{B}_\delta] + \mathbb{P}[\|B_n - B\| \geq \delta/2] \longrightarrow \mathbb{P}[B \in \mathcal{B}_\delta].\end{aligned}$$

Then we can take limit as $\delta \to 0$ to conclude. □

6.2 Convergence in Distribution

Definition 6.12. Given a sequence of random variables B_n and another random variable B, we say B_n **converges in distribution** to B if for all continuity points b of the cdf of B,

$$\mathbb{P}[B_n \leq b] \longrightarrow \mathbb{P}[B \leq b].$$

Example 6.13. Why the continuity caveat? Consider $B_n \sim \text{Uniform}[0, n^{-1}]$. We'd want to say $B_n \xrightarrow{\text{d}} 0$.

Theorem 6.14 (Lindeberg-Levy CLT)**.** *Let W_i be iid d_w-vectors with finite variance. Then,*

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} (W_i - \mathbb{E}[W_i]) \xrightarrow{\text{d}} \mathcal{N}(0, \text{Var}[W_i]).$$

Example 6.15 (Binning Estimator). Let $U_i := Y_i - \mu(X_i)$. Then $\mathbb{E}[U_i|X_i] = 0$, so $\mathbb{E}[U_i] = 0$. The binning estimator is given by

$$B_n(x) = \frac{\sum(\mu(X_i) + U_i)\mathbb{1}[X_i = x]}{\sum \mathbb{1}[X_i = x]} = \mu(x) + \frac{\sum U_i \mathbb{1}[X_i = x]}{\sum \mathbb{1}[X_i = x]}.$$

Thus,

$$\sqrt{n}(B_n(x) - \mu(x)) = \frac{\sqrt{n} n^{-1} \sum U_i \mathbb{1}\{X_i = x\}}{n^{-1} \sum \mathbb{1}\{X_i = x\}}.$$

We can apply the CLT to the numerator, and the WLLN to the denominator, then use Slutsky to combine.

Theorem 6.16 (Helly-Bray). *$B_n \xrightarrow{\text{d}} B$ if and only if $\mathbb{E}[f(B_n)] \to \mathbb{E}[f(B)]$ for all bounded uniformly-continuous real-valued functions f.*

Proof. $\Rightarrow$: Take compact $\mathcal{B}$ so that $\mathbb{P}[B \notin \mathcal{B}] < \varepsilon$. Note that by compactness, f is uniformly continuous on $\mathcal{B}$. Then, partition $\mathcal{B}$ into cells $\mathcal{B}_j$ such that f varies by less than ε on $\mathcal{B}_j$. We can choose the boundaries of the cells to be continuity points of B. Thus we have $\mathbb{P}[B_n \in \mathcal{B}_j] \to \mathbb{P}[B \in \mathcal{B}_j]$.

Now let $f^*(b) := \sum_j \mathbb{1}\left\{b \in \mathcal{B}_j\right\} f(b_j^*)$ for some fied $b_j^* \in \mathcal{B}_j$. Then,

$$\begin{aligned} \mathbb{E}[f(B_n) - f(B)] &= \mathbb{E}[f(B_n) - f^*(B_n)] + \mathbb{E}[f^*(B_n) - f^*(B)] + \mathbb{E}[f^*(B) - f(B)] \\ &=: \mathrm{I} + \mathrm{II} + \mathrm{III}. \end{aligned}$$

Note that I and III are small because $f - f^* < \varepsilon$ on $\mathcal{B}$, $\mathbb{P}[B_n \notin \mathcal{B}] \to \mathbb{P}[B \notin \mathcal{B}] < \varepsilon$, and f is bounded. II is small because $\mathbb{P}[B_n \in \mathcal{B}_j] \to \mathbb{P}[B \in \mathcal{B}_j]$ and $f^*(b)$ is constant on each $\mathcal{B}_j$.

$\Leftarrow$: Note that $\Phi_n(b) = \mathbb{E}[\mathbb{1}\{B_n \leq b\}]$, where $\mathbb{1}\{\cdot \leq b\}$ is *almost*-continuous. We can approximate $\mathbb{1}\{\cdot \leq b\}$ by bounded uniformly-continuous functions. □

These functions can also be taken to be smooth.

It turns out that the following is also true:

Proposition 6.17 (Almost-Continuous Helly-Bray). *$B_n \xrightarrow{\text{d}} B$ if and only if $\mathbb{E}[f(B_n)] \to \mathbb{E}[f(B)]$ for all bounded real-valued functions f that are continuous on some $\mathcal{B}$ with $\mathbb{P}[B \in \mathcal{B}] = 1$.*

Proposition 6.18. *$B_n \xrightarrow{\text{p}} B$ implies $B_n \xrightarrow{\text{d}} B$, and if B is degenerate, then the converse also holds.*

Proof. $\Rightarrow$: From CMP, we have $f(B_n) \xrightarrow{\text{p}} f(B)$. Since f is bounded, we have $\mathbb{E}[f(B_n)] \to \mathbb{E}[f(B)]$.

$\Leftarrow$: Fix small $\varepsilon > 0$ and define $f(b) := \min\{\varepsilon, \|b - \beta\|\}$. Note that f is nonnegative, continuous, and bounded by ε. By Helly-Bray, we have $\mathbb{E}[f(B_n)] \to \mathbb{E}[f(B)] = 0$, so $0 \leq \mathbb{E}[f(B_n)] < \delta$ eventually for any $\delta > 0$. Taking $\delta \ll \varepsilon$, we can show $\mathbb{P}[|B_n - \beta| > \varepsilon] < \delta/\varepsilon$ eventually, so $B_n \xrightarrow{\text{p}} \beta$. □

Convergence in probability of a vector is equivalent to convergence in distribution of each component. Convergence in distribution of a vector implies convergence in distribution of each component by the continuous mapping theorem, but the converse does not hold, since the correlation structure can change by n. A partial converse can be shown for the case where the limit distribution is degenerate, using a similar argument as in the proof of the previous proposition, breaking the vector into components using LIE.

Proposition 6.19. *If $[A_n', B_n]' \xrightarrow{d} [A', \beta']'$, then $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{d} \beta$.*

Conversely, suppose that $B = \beta$ is constant. Then $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{d} \beta$ imply $[A_n', B_n']' \xrightarrow{d} [A', \beta']'$.

Theorem 6.20 (Continuous Mapping Theorem (in Distribution))**.** *If $B_n \xrightarrow{d} B$ and f is continuous (potentially vector-valued, potentially unbounded), then $f(B_n) \xrightarrow{d} f(B)$.*

Note that here we require f to be continuous everywhere, but we'll weaken this later.

Proof. Composition of bounded continuous function with f is still bounded and continuous, so we can apply Helly-Bray to conclude. $\square$

As a result of the almost-continuous version of Helly-Bray, we can weaken the continuity requirement in the CMT in distribution:

Proposition 6.21. *If $B_n \xrightarrow{d} B$ and f is continuous on some $\mathcal{B}$ with $\mathbb{P}[B \in \mathcal{B}] = 1$, then $f(B_n) \xrightarrow{d} f(B)$.*

Example 6.22. Given $\sqrt{n}(B_n - \beta) \xrightarrow{d} \mathcal{N}(0, \Sigma)$, we have

$$\sqrt{n}(cB_n - c\beta) \xrightarrow{d} \mathcal{N}(0, c\Sigma c')$$

and

$$\left\|(\Sigma/n)^{-1/2}(B_n - \beta)\right\|^2 \xrightarrow{d} \|\mathcal{N}(0, \mathrm{Id})\|^2 \sim \chi^2_{d_b}.$$

This is called **standardizing**. By contrast, **studentizing** is done by replacing Σ with a consistent estimator $\hat{\Sigma}$.

6.3 Slutsky's Theorem

Theorem 6.23. *Suppose $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{p} \beta$, where A, B can be vectors/matrices, and β is constant. Then,*

(i) $A_n + B_n \xrightarrow{d} A + \beta$.

(ii) $B_n A_n \xrightarrow{d} \beta A$.

(iii) $B_n^{-1} A_n \xrightarrow{d} \beta^{-1} A$ *as long as β is invertible.*

Proof. By a previous result, (A_n, B_n) jointly converge in distribution to A and β. Then (i) and (ii) follow from the continuous mapping theorem. (iii) follows similarly by observing that $B_n^{-1} \xrightarrow{p} \beta^{-1}$. $\square$

Example 6.24. Studentization; see previous example.

Proposition 6.25 (Delta Method)**.** *Suppose* $\sqrt{n}(B_n - \beta) \xrightarrow{d} \mathcal{N}(0, \Sigma)$. *If* f *is a vector-valued function that is continuously differentiable in a neighborhood of* β *with Jacobian* $\dot{f}(\beta)$, *then*

$$\sqrt{n}(f(B_n) - f(\beta)) \xrightarrow{d} \mathcal{N}\left(0, \dot{f}(\beta)\Sigma\dot{f}(\beta)'\right).$$

Proof. Denote the gradient of f_j at β by $\dot{f}_j(\beta)$. The mean-value form of a first-order Taylor expansion of $f_j(B_n)$ around β gives

$$\sqrt{n}\left(f_j(B_n) - f_j(\beta)\right) = \dot{f}_j(B_n^\star)\sqrt{n}(B_n - \beta),$$

where $B_n^\star$ is a convex combination of B_n and β. Since $B_n \to \beta$ we have $B_n^\star \xrightarrow{p} \beta$, so CMT gives $f_j(B_n^\star) \xrightarrow{p} \dot{f}_j(\beta)$. We stack over j and apply Slutsky's theorem to conclude. □

Note the danger in applying delta method. Functions that are "close" to being non-differentiable can have very different limit distributions. But even for smooth functions, a first-order approximation can be quite bad.

Example 6.26. Suppose $\sqrt{n}(B_n - \beta) \xrightarrow{d} \mathcal{N}(0, 1)$ and consider $f(b) = b^2$. Delta method says $\sqrt{n}(B_n^2 - \beta^2) \xrightarrow{d} \mathcal{N}(0, 4\beta^2)$ which is $\mathcal{N}(0, 0) = 0$ when $\beta = 0$! This comes from the fact that the first order approximation to f at $x = 0$ is just the zero function.

This is not useful. For $\beta \approx 0$ the approximation χ_1^2 is much more appropriate.

6.4 Proving the Central Limit Theorem

6.4.1 Proofs via Lindberg Swapping

Theorem 6.27 (Generic CLT)**.** *Suppose* $\{W_{ni}\}_{i=1}^n$ *are independent scalar random variables with* $\mathbb{E}[W_{ni}] = 0$ *and* $\mathrm{Var}[W_{ni}] = n^{-1}$. *Suppose that the* ***Lyapunov condition*** *holds, i.e.,* $\sum_{i=1}^n \mathbb{E}[|W_{ni}|^3] \to 0$ *as* $n \to \infty$. *Then,*

$$\sum_{i=1}^n W_{ni} \xrightarrow{d} \mathcal{N}(0, 1).$$

As a direct corollary, we have the following results:

Theorem 6.28 (Lyapunov's CLT)**.** *Suppose* $\{W_i\}_{i=1}^n$ *are independent, but* not *identically distributed scalar random variables. Suppose* $\mathbb{E}[|W_i|^3] \le c$ *for all* i *for some finite* c. *Then,*

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \frac{W_i - \mathbb{E}[W_i]}{\sqrt{\mathrm{Var}[W_i]}} \xrightarrow{d} \mathcal{N}(0, 1),$$

Theorem 6.29 (Lindeberg-Lévy Scalar CLT)**.** *Suppose* $\{W_i\}_{i=1}^n$ *are iid scalar random variables with finite variance but not necessarily any third moment bound. Then,*

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n \frac{(W_i - \mathbb{E}[W_i])}{\sqrt{\mathrm{Var}[W_i]}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Proof. Truncation argument. □

6.4.2 Proof via Characteristic Functions

The motivation is to extend CLT to the multivariate case, but multivariate Taylor series expansion is a mess.

Definition 6.30. For a random vector B,

- the **moment generating function** is $t \mapsto \mathbb{E}[\exp(t'B)]$,
- the **characteristic function** is $\phi : t \mapsto \mathbb{E}[\exp(\mathrm{i}t'B)]$.

Note that the moment generating function is not guaranteed to be finite, but the characteristic function is always well-defined and bounded by 1.

Theorem 6.31 (Lévy's Continuity Theorem)**.** *$B_n \xrightarrow{\mathrm{d}} B$ if and only if $\mathbb{E}[\exp(\mathrm{i}(t'B_n))] \to \mathbb{E}[\exp(\mathrm{i}(t'B))]$ for all t, i.e., we have pointwise convergence of the characteristic functions.*

This can be read as another refinement of Helly-Bray.

Proof. $\Rightarrow$ is straightforward. For $\Leftarrow$, consider multivariate normal A. Pointwise convergence of characteristic functions gives convergence of densities of $B_n + \sigma A$ to that of $B + \sigma A$ for all $\sigma > 0$, which gives

$$\mathbb{E}[f(B_n + \sigma A)] \to \mathbb{E}[f(B + \sigma A)]$$

for all bounded smooth f. Thus $B_n + \sigma A \xrightarrow{\mathrm{d}} B + \sigma A$ for all $\sigma > 0$, which implies $B_n \xrightarrow{\mathrm{d}} B$. ☐

We have another refinement of Helly-Bray:

Theorem 6.32 (Cramér-Wold Device)**.** *$B_n \xrightarrow{\mathrm{d}} B$ if and only if $t'B_n \xrightarrow{\mathrm{d}} t'B$ for every t.*

That is, a probability measure on $\mathbb{R}^d$ is uniquely determined by the totality of its one-dimensional projections.

Proof. Comes directly from Lévy's continuity theorem. ☐

This gives the following version of a multivariate CLT:

Theorem 6.33 (Lindeberg-Lévy CLT)**.** *Let W_i be iid d_w-vectors with finite variance. Then,*

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} (W_i - \mathbb{E}[W_i]) \xrightarrow{\mathrm{d}} \mathcal{N}(0, \mathrm{Var}[W_i]).$$

Proof. Use the Cramer-Wold device to reduce to the scalar case. ☐

7 Instrumental Variables

Motivation: we think of D_i as endogenous, so selection on observables misses the point, and common trends assumes unobservable confounders are time-invariant.

Setup:

- Treatment variable D_i, potentially multivalued or a vector.
- Instrument variables Z_i, **instruments** for short.
- Outcome Y_i. Potential outcomes $Y_i(d, z)$ later assumed to depend only on d.
- Other observable variables (covariates) W_i.

All IV models will be based roughly on the following assumptions:

(i) **Exclusion**: $Y_i(d, z) = Y_i(d)$ for all d, z,

(ii) **Exogeneity**: $Y_i(d, z)$ and Z_i are independent conditional on W_i for all d,

(iii) **Relevance**: Z_i and D_i are dependent (in an appropriate sense) conditional on W_i,

where the precise meanings of (ii) and (iii) depend on the specifics of the IV model. The idea of the IV strategy is to allow D_i and $Y_i(d)$ to be dependent and shift D_i indirectly with Z_i.

7.1 The Linear IV Model

We will mainly discuss the **linear IV** model:

Assumption 7.1 (Linear IV).

- **Exclusion**: $Y_i(d, z) = Y_i(d)$ for all d, z.
- **Exogeneity**: $Y_i(d) \perp\!\!\!\perp Z_i | W_i$ for all d.
- **Linear functional form**: There is a *known* vector-valued function g such that

$$Y_i(d) = g(d, W_i)'\beta + U_i, \quad \mathbb{E}[U_i|W_i] = 0.$$

- **Relevance**: will depend on the specific setting and be made precise later.

Remark 7.2.

- Exogeneity and linear functional form gives $\mathbb{E}[U_i|Z_i, W_i] = \mathbb{E}[U_i|W_i] = 0$.
- Linear functional form restricts U_i to not depend on d. As a consequence, it implicitly assumes **constant treatment effects**:

$$Y_i(d_1) - Y_i(d_0) = \left[g(d_1, W_i) - g(d_0, W_i)\right]' \beta.$$

  Treatment effect is assumed to be linear in parameters and does not depend on unobservables. We will address the limitations of this assumption later.
- Linear regression is nested as a special case with $Z_i = D_i$. In this sense IV generalizes linear regression by allowing for endogeneity.

With the linear IV assumptions and writing $X_i := g(D_i, W_i)$, we have

$$Y_i = X_i'\beta + U_i, \qquad \mathbb{E}[U_i|Z_i, W_i] = 0.$$

Two weaker implications typically sufficient for the statistical theory are $\mathbb{E}[U_i Z_i] = 0$ and $\mathbb{E}[U_i W_i] = 0$. Note, however, that they are true for different reasons: The first equality comes from the exogeneity assumption, while the second comes from g being correctly specified. Most of the theory will just depend on these two orthogonality conditions, however they are justified.

Components of X_i are traditionally divided into the following two types: those that depend on D_i (and potentially also on W_i) are called **endogenous regressors**, while those that depend *only* on W_i are called **exogenous regressors**. The endogenous regressor label is fine if we take it to mean mean dependence with $Y_i(d)$. The exogenous regressor label is more problematic, since W_i often include variables like sociodemographic controls that are correlated with $Y_i(d)$. **Control variables** or **covariates** is probably a better name.

7.2 Identification in the Linear Model

We will work with increasingly general settings.

7.2.1 Simple Linear IV

Consider first the case where $X_i = [D_i, 1]'$ with D_i a scalar, so $W_i = 1$.

Assumption 7.3 (Relevance for Simple Linear IV). $\operatorname{Cov}[D_i, Z_i] \neq 0$.

By exogeneity,

$$\operatorname{Cov}[Y_i, Z_i] = \operatorname{Cov}[\beta_1 D_i + \beta_2 + U_i, Z_i] = \beta_1 \operatorname{Cov}[D_i, Z_i].$$

If relevance is satisfied, we can identify β_1 as

$$\beta_1 = \frac{\operatorname{Cov}[Y_i, Z_i]}{\operatorname{Cov}[D_i, Z_i]} = \frac{\operatorname{Cov}[Y_i, Z_i] / \operatorname{Var}[Z_i]}{\operatorname{Cov}[D_i, Z_i] / \operatorname{Var}[Z_i]} = \frac{\text{“reduced form”}}{\text{“first stage’}}.$$

Thus β_1 can be interpreted as the ratio of two linear regressions, with the reduced form being the causal effect of Z_i on Y_i not accounting for how much D_i changes, and the division by the first stage adjusting for how much D_i changes as Z_i changes.

7.2.2 Identification with Covariates

Let $X_i = [D_i, W_i']'$ and write the **first stage variables** as $F_i := [Z_i, W_i']'$.

Assumption 7.4 (Relevance with Covariates). $\mathbb{E}[F_i X_i']$ is invertible.

Under relevance, we have:

Proposition 7.5. $\mathbb{E}[F_i F_i']$ *is invertible.*

Proof. The invertibility of $\mathbb{E}[F_i X_i']$ implies that there is no perfect collinearity in F_i (otherwise the rows of $\mathbb{E}[F_i X_i']$ aren't linearly independent), which implies that $\mathbb{E}[F_i F_i']$ is invertible. $\square$

We can use the same identification logic, this time pre-multplying by F_i:

$$\mathbb{E}[F_i Y_i] = \mathbb{E}[F_i X_i']\beta + \mathbb{E}[F_i U_i] = \mathbb{E}[F_i X_i']\beta.$$

Then, if relevance is satisfied, we have

$$\beta = \mathbb{E}[F_i X_i']^{-1}\,\mathbb{E}[F_i Y_i] = \underbrace{\left(\mathbb{E}[F_i F_i']^{-1}\,\mathbb{E}[F_i X_i']\right)}_{\text{first stage}}{}^{-1}\;\underbrace{\left(\mathbb{E}[F_i F_i']^{-1}\,\mathbb{E}[F_i Y_i]\right)}_{\text{reduced form}}.$$

Interpreting the relevance condition: Note that relevance holds if and only if the first stage coefficient matrix is invertible:

$$\mathbb{E}[F_i F_i']^{-1}\,\mathbb{E}[F_i X_i'] = \begin{bmatrix} \gamma_1 & 0 & \dots & \dots & 0 \\ \gamma_2 & 1 & 0 & \dots & 0 \\ \vdots & 0 & 1 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma_{d_x} & 0 & \dots & \dots & 1 \end{bmatrix}.$$

Thus relevance holds if and only if a regression of D_i onto $[Z_i, W_i']$ has non-zero coefficient on Z_i, that is, if D_i and Z_i are correlated after projecting out W_i.

Residualized RF/FS Interpretation: Write $D_i = \gamma_1 Z_i + \gamma_2' W_i + V_i$, where $\mathbb{E}[V_i F_i] = 0$. Substituting into the outcome equation gives

$$\begin{aligned} Y_i &= \beta_1 D_i + W_i'\beta_2 + U_i \\ &= (\beta_1\gamma_1) Z_i + W_i'(\beta_1\gamma_2 + \beta_2) + (\beta_1 V_i + U_i). \end{aligned}$$

One can verify that $\mathbb{E}[Z_i(\beta_1 V_i + U_i)] = 0$ and $\mathbb{E}[W_i(\beta_1 V_i + U_i)] = 0$, so writing $\tilde{Z}_i$ for the residual of a regression of Z_i on W_i, we have

$$\beta_1 = \frac{\beta_1\gamma_1}{\beta_1} = \frac{\mathrm{Cov}[Y_i, \tilde{Z}_i]/\,\mathrm{Var}[\tilde{Z}_i]}{\mathrm{Cov}[D_i, \tilde{Z}_i]/\,\mathrm{Var}[\tilde{Z}_i]} = \frac{\text{residualized RF}}{\text{residualized FS}}.$$

7.2.3 Vector Endogenous Variables

Let's now extend the analysis to most general case, with vector endogenous variables. This case can arise with different endogenous variables, nonlinear function forms (D_i, D_i^2, D_i^3), or accounting for observed heterogeneity through interactions with observables $(D_i, D_i W_{i1})$.

Again writing $X_i = [D_i', W_i']'$ and $F_i = [Z_i', W_i']'$, we have

$$\mathbb{E}[F_i U_i] = 0 \iff \mathbb{E}[F_i Y_i] = \mathbb{E}[F_i X_i']\beta.$$

Thus to have a solution, we need the **order condition** $d_f \geq d_x$. Since X_i and F_i contain the same subvector W_i, we need at least $d_x - d_w$ **excluded variables** in F_i — components of F_i that are not in X_i — to have hope of identification.

One usually generate these additional excluded variables by mirroring the vector endogenous variables, taking for instance Z_i, Z_i^2, Z_i^3 in the nonlinear case and $Z_i, Z_i W_{i1}$

in the interaction case. In principle one can use any function of Z_i as an instrument, since the orthogonality condition implies

$$\mathbb{E}\left[h(W_i,Z_i)U_i\right] = \mathbb{E}\Big[\,\mathbb{E}\left[h(W_i,Z_i)U_i|W_i,Z_i\right]\Big] = 0, \quad \forall h,$$

but anything nonstandard will smell bad.

7.2.4 Identification in the General Case

In the **exactly identified** case $d_f = d_x$, the relevance condition is that $\mathbb{E}[F_iX_i']$ is invertible, and as before we have under relevance that

$$\beta = \mathbb{E}[F_iX_i']^{-1}\,\mathbb{E}[F_iY_i].$$

In the **overidentified** case $d_f > d_x$, the relevance condition becomes $\mathbb{E}[cF_iX_i']$ is invertible for some deterministic $d_x \times d_f$ matrix c. When relevance is satisfied, we have

$$\beta = \mathbb{E}[cF_iX_i']^{-1}\,\mathbb{E}[cF_iY_i].$$

This embeds the exactly identified case, with c cancelling out in the expression above.

One can think of the overidentified case as having multiple possible excluded vectors cF_i. Under the maintained assumptions, different c's will give the same β. This is a testable implication of the model, and leads to the **overidentification test**.

The following is easily established:

Proposition 7.6. *Let C be a consistent estimator of c and assume the maintained assumptions hold. Then,*

$$B_{\text{IV}}(c) := \left(C\sum_i F_iX_i'\right)^{-1} C\left(\sum_i F_iY_i\right)$$

is consistent for β and

$$\sqrt{n}(B_{\text{IV}}(c) - \beta) \xrightarrow{\text{d}} \mathcal{N}(0, \Sigma(c)),$$

where

$$\Sigma(c) := \left(c\,\mathbb{E}[F_iX_i']\right)^{-1} c\,\mathbb{E}[U_i^2F_iF_i']c'\,\left(\mathbb{E}[X_iF_i']c'\right)^{-1}.$$

The asymptotic variance can be estimated in the usual way, with $\hat{U}_i := Y_i - X_i'B_{\text{IV}}(c)$.

Remark 7.7. To get some intuition for the variance, let's consider the simple linear IV case with $X_i = [D_i, 1]'$ and $F_i = [Z_i, 1]'$, where we have

$$\Sigma_{\text{IV}} = \mathbb{E}\left[U_i^2(\mathbb{E}[F_iX_i']^{-1}F_i)(\mathbb{E}[F_iX_i']^{-1}F_i)'\right].$$

Assuming homoscedasticity in the instrument $\mathbb{E}[U_i^2|Z_i] = \mathbb{E}[U_i^2]$, one can show that the first diagonal term of Σ_{IV} is

$$\frac{\text{Var}[U_i]}{\text{Corr}[D_i,Z_i]^2\,\text{Var}[D_i]}.$$

This has the same structure as in case with the simple linear regression, except in this case the signal is $\text{Corr}[D_i,Z_i]^2\,\text{Var}[D_i] \le \text{Var}[D_i]$. ☕

7.3 Two-Stage Least Squares

It is natural to ask what the optimal c is. The intuition above suggests choosing c so that cF_i are highly correlated with X_i. Thus consider

$$c_{\text{TSLS}} \coloneqq \mathbb{E}[X_i F_i'] \, \mathbb{E}[F_i F_i']^{-1} = \begin{bmatrix} \mathbb{E}[X_{i1} F_i'] \, \mathbb{E}[F_i F_i']^{-1} \\ \vdots \\ \mathbb{E}[X_{id_x} F_i'] \, \mathbb{E}[F_i F_i'] \end{bmatrix} = \begin{bmatrix} \text{coef. } X_{i1} \text{ on } F_i{}' \\ \vdots \\ \text{coef. } X_{id_x} \text{ on } F_i{}' \end{bmatrix},$$

so that $cF_i = \mathbb{L}[X_i|F_i]$. With this choice of c, we have

$$\begin{aligned} \beta &= \mathbb{E}[c_{\text{TSLS}} F_i X_i']^{-1} \, \mathbb{E}[c_{\text{TSLS}} F_i Y_i] \\ &= \mathbb{E}\left[\mathbb{L}[X_i|F_i] X_i'\right]^{-1} \mathbb{E}\left[\mathbb{L}[X_i|F_i] Y_i\right] \\ &= \mathbb{E}\left[\mathbb{L}[X_i|F_i] \mathbb{L}[X_i|F_i]'\right]^{-1} \mathbb{E}\left[\mathbb{L}[X_i|F_i] Y_i\right], \end{aligned}$$

where the last equality comes from the orthogonality of $X_i - \mathbb{L}[X_i|F_i]$ and $\mathbb{L}[X_i|F_i]$. This is the **two-stage least squares** estimator, where the name comes from the interpretation of first regressing X_i onto F_i to get $\mathbb{L}[X_i|F_i]$, and then regress Y_i onto $\mathbb{L}[X_i|F_i]$.

For $c = c_{\text{TSLS}}$, the relevance condition is the invertibility of

$$\mathbb{E}[c_{\text{TSLS}} F_i X_i'] = \mathbb{E}[X_i F_i'] \, \mathbb{E}[F_i F_i']^{-1} \, \mathbb{E}[F_i X_i'].$$

Assuming that there is no perfect collinearity in F_i, relevance holds if and only if the first stage coefficients matrix $\mathbb{E}[X_i F_i'] \, \mathbb{E}[F_i F_i']^{-1}$ has full rank. If we organize X_i and F_i so that W_i are placed at the end, then

$$\mathbb{E}[F_i F_i']^{-1} \, \mathbb{E}[F_i X_i'] = \begin{bmatrix} \gamma_1 & 0_{(d_x - d_w) \times d_w} \\ \gamma_2 & \mathrm{Id}_{d_w} \end{bmatrix}$$

has full rank if and only if γ_1 has full rank. A necessary condition for this is that each endogenous variable can be shifted by some excluded variable after controlling for covariates.

The sample analog of c_{TSLS} is $C_{\text{TSLS}} \coloneqq X'F(F'F)^{-1}$, which gives

$$\begin{aligned} B_{\text{TSLS}} \equiv B_{\text{IV}}(c_{\text{TSLS}}) &\coloneqq \left(C_{\text{TSLS}} \sum_i F_i X_i' \right)^{-1} C_{\text{TSLS}} \left(\sum_i F_i Y_i \right) \\ &= \left(X'F(F'F)^{-1}F'X \right)^{-1} X'F(F'F)^{-1}F'Y \\ &= (X'P_F X)^{-1} X' P_F Y = \left((P_F X)'(P_F X) \right)^{-1} (P_F X)'Y. \end{aligned}$$

7.4 Generalized Method of Moments Estimation

As one might expect, the TSLS estimator is efficient under the homoscedasticity case, but not in general. The general case is handled by the **generalized method of moments** (GMM) estimator $B_{\text{IV}}(c^\star)$:

Proposition 7.8 (Efficiency of GMM). *Let* $c^\star := \mathbb{E}[X_i F_i']\Omega^{-1}$, *where* $\Omega := \mathbb{E}[U_i^2 F_i F_i']$. *Then* $c^\star$ *is optimal in the sense that* $\Sigma_{IV}(c) - \Sigma_{IV}(c^\star)$ *is positive semidefinite for all* c.

Proof *(Sketch)*. Write $q := \mathbb{E}[X_i F_i']$, so that

$$\Sigma_{\text{IV}}(c) = (cq')^{-1} c\Omega c' (qc')^{-1}, \quad \Sigma_{\text{IV}}(c^\star) = (q\Omega^{-1}q')^{-1}.$$

One can then show $\Omega_{\text{IV}}(c) - \Sigma_{\text{IV}}(c^\star) = aa'$ for some a and thus is positive semidefinite. □

Corollary 7.9 (Efficiency of TSLS). *If* $\mathbb{E}[U_i^2|F_i] = \mathbb{E}[U_i^2]$, *then* $c^\star = \mathbb{E}[U_i^2]c_{TSLS}$. *Thus TSLS is the efficient weighting under homoscedasticity.*

The optimal (two step) GMM estimator is then given by

$$B_{\text{OGMM}} \equiv B_{\text{IV}}(C^\star),$$

where

$$C^\star := \left(\sum_i X_i F_i'\right)\left(\sum_i \hat{U}_i^2 F_i F_i'\right)^{-1}, \quad \hat{U}_i := Y_i - X_i' B_{\text{TSLS}}.$$

The folklore is that B_{OGMM} can behave considerable worse than B_{TSLS} in finite samples due to the difficulty of estimating a fourth moment object $\mathbb{E}[U_i^2 F_i F_i']$ in contrast to the second moment object need in TSLS $\mathbb{E}[X_i F_i']$. Thus for linear models, the common practice is to use B_{TSLS} even though B_{OGMM} is more efficient.

8 IV Statistics

8.1 The Weak Instruments Problem

Recall the interpretation of the IV estimator as the ratio of the reduced form and first stage coefficients. When the first stage coefficients are small, the distribution of the IV estimator can deviate significantly from normality in finite samples (even though it is asymptotically normal).

One way to see the weak instruments problem is through simulation. The Nelson and Startz (1990) simulation uses the following data generating process:

- $Y_i = 0 \cdot D_i + U_i = U_i$ (i.e., $\beta = 0$).
- $D_i = \gamma Z_i + V_i$, where the value of γ will be varied to simulate different degrees of instrument strength.
- (U_i, V_i) is bivariate normal with a correlation of $\rho = .99$ and independent of Z_i, which is also standard normal.

One can see from simulation the nonnormality of the IV estimator when γ is small. Let's consider the distribution when $\gamma = 0$. Note that

$$B_{\text{IV}} - \beta = \frac{\frac{1}{\sqrt{n}} \sum_i Z_i U_i}{\frac{1}{\sqrt{n}} \sum_i Z_i D_i},$$

$$\frac{1}{\sqrt{n}} \sum_i \begin{bmatrix} Z_i U_i \\ Z_i D_i \end{bmatrix} \xrightarrow{\text{d}} \mathcal{N}\left(0, \begin{bmatrix} \text{Var}[Z_i U_i] & \text{Cov}[Z_i U_i, Z_i D_i \\ \text{Cov}[Z_i U_i, Z_i D_i] & \text{Var}[Z_i D_i] \end{bmatrix}\right).$$

It follows that

$$B_{\text{IV}} - \beta \xrightarrow{\text{d}} \frac{\text{Var}[Z_i U_i] \cdot Q}{\text{Var}[Z_i D_i] \cdot R}, \qquad \begin{bmatrix} Q \\ R \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right)$$

with $\rho := \text{Corr}[Z_i U_i, Z_i D_i]$. We can decompose $Q = \rho R + (1 - \rho^2)^{1/2} P$, where $P \sim \mathcal{N}(0, 1)$ is independent of R, to get

$$B_{\text{IV}} - \beta \xrightarrow{\text{d}} \underbrace{\rho \left(\frac{\text{Var}[Z_i U_i]}{\text{Var}[Z_i D_i]}\right)}_{\text{constant shift}} + \left((1 - \rho^2) \cdot \frac{\text{Var}[Z_i U_i]}{\text{Var}[Z_i D_i]}\right) \frac{P}{R}.$$

Under homoscedasticity, one can show with some algebra that the constant shift is equal to $\beta_{\text{OLS}} - \beta$. Thus

$$B_{\text{IV}} \xrightarrow{\text{d}} \beta_{\text{OLS}} + \left((1 - \rho^2) \cdot \frac{\text{Var}[Z_i U_i]}{\text{Var}[Z_i D_i]}\right) \cdot \text{Cauchy distribution.}$$

This discontinuous limit theory — with the limiting distribution being normal for $\gamma \neq 0$ and Cauchy when $\gamma = 0$ — suggests that something might break near 0, and indeed this is the case shown in simulation.

Note of course that the asymptotic normality result still holds for $\gamma \neq 0$, the problem is only that we may need an extremely large n for the approximation to be good.

8.2 Weak Instrument Asymptotics

An approach to smooth out the asymptotic theory is a **local asymptotics** framework. Consider a sequence of data generating processes indexed by n:

$$D_i = \gamma_n Z_i + V_i, \qquad \gamma_n := \frac{\gamma}{\sqrt{n}}.$$

This is not meant to reflect how the actual DGP changes as the number of observations increases. Rather, it is only a mental experiment that leads to a different asymptotic approximation of B_{IV}. An actual DGP $D_i = \tilde{\gamma} Z_i + V_i$ giving n observations is thought of as arising in the nth slice of this sequence of DGP, with $\gamma_n = \tilde{\gamma}$ and thus $\gamma = \sqrt{n}\tilde{\gamma}$.

The same argument as before gives

$$\begin{aligned} B_{\text{IV}} - \beta &= \frac{\frac{1}{\sqrt{n}} \sum_i Z_i U_i}{\frac{1}{\sqrt{n}} \sum_i Z_i D_i} = \frac{\frac{1}{\sqrt{n}} \sum_i Z_i U_i}{\sqrt{n}\gamma_n \cdot \frac{1}{n} \sum_i Z_i^2 + \frac{1}{\sqrt{n}} \sum_i Z_i V_i} \\ &\xrightarrow{\text{d}} \frac{\operatorname{Var}[Z_i U_i]^{1/2} Q}{\gamma\, \mathbb{E}[Z_i^2] + \operatorname{Var}[Z_i V_i]^{1/2} R}, \end{aligned}$$

where Q, R is a bivariate normal with correlation ρ.

Thus, given that the actual data generating process is $D_i = \tilde{\gamma} Z_i + V_i$, the distribution above with $\gamma = \sqrt{n}\tilde{\gamma}$ is an asymptotic approximation of $B_{\text{IV}} - \beta$. Importantly, this asymptotic distribution is continuous in $\tilde{\gamma}$. With $\tilde{\gamma} = 0$, this reduces to the Cauchy case, and with $\gamma = \sqrt{n}\tilde{\gamma} \gg 0$, $\operatorname{Var}[Z_i V_i]^{1/2} R$ is negligible and we approach the normal distribution.

This approximation turns out to perform much better in simulation. However, it relies on us knowing $\gamma = \sqrt{n}\tilde{\gamma}$, which we don't really know how to estimate — $\sqrt{n}C$, with C the first stage OLS estimator for $\tilde{\gamma}$, is not consistent for γ.[6]

8.3 Weak Instrument Diagonstics

Even though γ cannot be consistently estimated, we can still construct asymptotically valid tests of hypotheses about γ to diagnose the strength of the instrument. We will formulate these tests in terms of the **first stage F-statistic**, which corresponds to the F-statistic for testing the null of $\gamma = 0$ in the first stage regression of D_i onto Z_i.

We assume there are no covariates (or they are all partialled out[7]) and denote the OLS estimator of D_i onto Z_i as C. Assume also homoscedasticity of first stage residual V_i, $\mathbb{E}[V_i^2 | Z_i] = \mathbb{E}[V_i^2] =: \sigma_V^2$. In the scalar case, we have

$$T_{\text{FSFS}} := \left(\frac{\sqrt{n} C}{V_{\text{FS,HOM}}^{1/2}} \right)^2 \xrightarrow{\text{d}} \left(\frac{\gamma\, \mathbb{E}[Z_i^2]^{1/2}}{\sigma_V} + R \right)^2 =: (\mu + R)^2.$$

where $V_{\text{FS,HOM}}$ denotes the homoskedastic estimator of the asymptotic variance of C and the convergence comes from the weak instrument asymptotics. To see the convergence,

[6] $\sqrt{n}$ is increasing at least as quickly as any reasonable estimator of $\tilde{\gamma}$ would be decreasing.

[7] With covariates, the FSFS still correspond to the null hypothesis that the coefficient on Z_i is zero, leaving the coefficients on the covariates unrestricted.

note that with the DGP considered in the weak instruments asymptotics, we have

$$\sqrt{n}C = \frac{\frac{1}{\sqrt{n}}\sum_i Z_i(Z_i\gamma_n + V_i)}{\frac{1}{n}\sum Z_i^2} = \sqrt{n}\gamma_n + \frac{\frac{1}{\sqrt{n}}\sum Z_i V_i}{\frac{1}{n}\sum Z_i^2} \xrightarrow{\text{d}} \gamma + \frac{\sigma_V R}{\mathbb{E}[Z_i^2]^{1/2}},$$

and for the denominator,

$$V_{\text{FS,HOM}}^{1/2} := \left(\frac{\frac{1}{n-1}\sum_i (D_i - Z_i C)^2}{\frac{1}{n}\sum_i Z_i^2}\right)^{1/2} \xrightarrow{\text{p}} \frac{\sigma_V}{\mathbb{E}[Z_i^2]^{1/2}}.$$

If Z_i is a vector, we similarly have

$$T_{\text{FSFS}} := \frac{1}{d_z}\left\|\sqrt{n}V_{\text{HOM}}^{-1/2}C\right\|^2 \xrightarrow{\text{d}} \frac{1}{d_z}\left\|\frac{\mathbb{E}[Z_i Z_i']^{1/2}\gamma}{\sigma_V} + R\right\| =: \frac{1}{d_z}\|\mu + R\|^2,$$

where R is as before a standard normal. The limiting distribution of $d_z T_{\text{FSFS}}$ — the distribution of $\|\mu + R\|^2$ — is called a **non-central chi-squared** distribution with d_z degrees of freedom and non-centrality parameter μ^2. We call $\mu^2 := \|\mu\|^2$ the **concentration parameter**. Note that the concentration parameter is a unit-free measure of how much of the variation in D_i is driven by variation in γZ_i.

By symmetry, the non-central chi-squared distribution depends on μ only through μ^2, with larger values of μ^2 monotonically increasing the quantiles of the distribution. Thus we may test $H_0 : \mu^2 \leq \bar{\mu}^2$ using T_{FSFS} as a test statistic and the quantiles of $d_z^{-1}\|\bar{\mu} + R\|^2$ as critical values. Specifically, we reject if $T_{\text{FSFS}} > c_\alpha$, where c_α is the $1 - \alpha$ quantile of the distribution of $d_z^{-1}\|\bar{\mu} + R\|^2$.

It remains to specify the cutoff $\bar{\mu}^2$.

8.3.1 Weak Instrument Definition #1: Bias

The natural idea is to start from our motivation for diagnosing weak instruments. We'd want to declare instruments as weak if the bias of B_{IV} is large. However, the bias is not always well-defined. In particular, when $d_z = 1$, $\mathbb{E}[B_{\text{IV}}]$ does not even exist due to Cauchy-ish fat tails.

Stock and Yogo (2005) considers the TSLS estimator with $d_z \geq 3$. With the same weak IV asymptotics and a bit more work, one can show that

$$B_{\text{TSLS}} - \beta \xrightarrow{\text{d}} \frac{\sigma_U}{\sigma_V}\left(\frac{(\mu + R)'(\rho R + (1-\rho^2)^{1/2}P)}{(\mu + R)'(\mu + R)}\right),$$

where $[R', P']'$ is a $2d_z$-dimensional standard normal random vector. Since $\mathbb{E}[P|R] = 0$, we have

$$\mathbb{E}[B_{\text{TSLS}} - \beta] = \frac{\sigma_U}{\sigma_V}\rho\,\mathbb{E}\left[\frac{(\mu + R)'R}{(\mu + R)'(\mu + R)}\right].$$

This depends not just on μ (which we know how to test), but also on σ_U, σ_V, and ρ. However, recall that under homoscedasticity, $B_{\text{OLS}} - \beta = (\sigma_U/\sigma_V)\rho$, so the **relative bias**

$$\text{relative bias} := \left|\frac{\mathbb{E}[B_{\text{TSLS}} - \beta]}{B_{\text{OLS}} - \beta}\right| = \left|\mathbb{E}\left[\frac{(\mu + R)'R}{(\mu + R)'(\mu + R)}\right]\right|$$

depends only on μ and in fact only on μ^2. This is good — we know how to do inference on μ via T_{FSFS}! One can calculate the expected bias as a function of μ^2 and d_z, and choose the cutoff $\bar{\mu}^2$ base on their taste for the relative bias. Stock and Yogo (2005) suggests 10% as the threshold for relative bias, which at a level of $\alpha = .05$ gives a critical value of around 10 for T_{FSFS} across many different values of d_z. This is the birth of the **"F-stat rule of thumb."**

8.3.2 Weak Instrument Definition #2: Nagar Bias

A rather weird idea.

8.3.3 Weak Instrument Definition #3: Size Distortion

We now consider the $d_z = 1$ setting. The idea (still proposed by Stock and Yogo (2005)) is to shift focus from estimation to inference. What is the *actual* size of a nominal level 5% t-test for $H_0 : \beta = \beta_0$?

$$\text{actual size} \coloneqq \underbrace{\mathbb{P}[T_{\text{IV}}(\beta) > 1.96]}_{\text{want to be .05 when } \beta = \beta_0}, \qquad T_{\text{IV}}(\beta_0) \coloneqq \left| \frac{B_{\text{IV}} - \beta_0}{(V_{\text{IV,HOM}}/n)^{1/2}} \right|.$$

It turns out that one can show using weak instrument asymptotics that

$$T_{\text{IV}}(\beta_0) \xrightarrow{\text{d}} \left| \frac{Q}{\left(1 - \frac{2\rho Q}{\mu + R} + \frac{Q^2}{(\mu + R)^2}\right)^{1/2}} \right|.$$

Note that as we send $\mu \to \infty$, the distribution converges to $|\mathcal{N}(0,1)|$, giving the correct size. But in general, the limiting distribution depends on μ and ρ.

One can pick a desired maximum size distortion level and declare instruments as weak if the size distortion is above that level. The problem is that we only how to test hypotheses about μ^2 and not ρ. Stock and Yogo (2005) proposes to consider the worst-case rejection over all ρ, and suggest a maximum size distortion of 15%, leading again to a T_{FSFS} thresholds of $9.05 \approx 10$.

However, the worse-case size distortion across ρ is achieved at $\rho = \pm 1$, which is a very extreme case. Angrist and Kolesár (2024) show that there is no size distortion for $|\rho| <\approx .565$. The intuition for this is that coverage is a race between bias and variance of estimates. With small μ, both are inflated. But bias is also controlled by $|\rho|$, and with $|\rho|$ small, variance wins.

8.4 Inference Robust to Weak Instruments

8.4.1 The Anderson-Rubin (1949) Test

Setup:

- $F_i = [Z_i', W_i']'$, $X_i = [D_i', W_i']'$.
- $Y_i = X_i'\beta + U_i$, $\mathbb{E}[U_i F_i] = 0$.

We hope to test $H_0 : \beta = \beta_0$. The idea is to define $U_i(\beta_0) \coloneqq Y_i - X_i' \beta_0$. Under H_0, we have $U_i(\beta_0) = U_i$, so

$$\alpha(\beta_0) \coloneqq \mathbb{E}[F_i F_i']^{-1}\, \mathbb{E}[F_i U_i(\beta_0)] = 0.$$

Note that $\alpha(\beta_0)$ is the coefficient from regressing $U_i(\beta_0)$ onto F_i. One can check easily that the converse is also true: $\alpha(\beta_0) = 0$ implies H_0. Thus, testing $\beta = \beta_0$ is equivalent to testing $\alpha_0 = 0$, which can be done through an F-test. Note that in the case $\beta_0 = 0$, we are simply testing the hypothesis that the reduced form coefficients are zero.

In some sense, the Anderson-Rubin test loads the weak instrument problem onto the power instead of the size of the test. To see this, note that

$$\alpha(\beta_0) = \mathbb{E}[F_i F_i']^{-1}\, \mathbb{E}[F_i(U_i + X_i'(\beta - \beta_0))] = \underbrace{\mathbb{E}[F_i F_i']^{-1}\, \mathbb{E}[F_i X_i']}_{\text{first stage coefficients}}(\beta - \beta_0).$$

So if relevance fails, $\mathbb{E}[F_i X_i']$ is rank deficient, and thus it can be the case that $\alpha(\beta_0) = 0$ even when $\beta \neq \beta_0$. As relevance becomes closer to being true, it becomes easier to determine when $\beta \neq \beta_0$. In the overidentified case ($d_f > d_x$), the AR test can have poor power since we are testing both $\beta = \beta_0$ and $\mathbb{E}[F_i U_i] = 0$. Some power is wasted on testing the latter.

8.4.2 Subvector AR Tests

We are usually not interested in testing the entire β vector. Suppose $X_i = [D_i', W_i']'$ with $\beta = [\beta_1', \beta_2']'$ and $F_i \coloneqq [Z_i', W_i']'$. In this case we probably only care about testing $H_0 : \beta_1 = \beta_{1,0}$.

To implement the AR test, we define

$$U_i(\beta_{1,0}) \coloneqq Y_i - D_i' \beta_{1,0}.$$

Under H_0, we have

$$\begin{aligned}
\mathbb{L}[U_i(\beta_{1,0})|F_i] &= \mathbb{L}[Y_i - D_i'\beta_{1,0}|F_i] \\
&= \mathbb{L}[W_i'\beta_2 + U_i|F_i] = Z_i'0 + W_i'\beta_2.
\end{aligned}$$

Thus we may regress $U_i(\beta_{1,0})$ onto F_i and test that the coefficient on Z_i is 0.

8.4.3 The Anderson-Rubin Confidence Intervals

To get a confidence interval, we can use the duality between confidence intervals and hypothesis tests:

$$C \coloneqq \{\beta_0 : \text{ AR test does not reject } H_0 : \beta = \beta_0\}.$$

Note that this is not necessarily symmetric nor necessarily bounded. In fact, one can show that C is unbounded if and only if one cannot reject H_0 : relevance with a F test.

9 Instrumental Variables: Heterogeneous Treatment Effects

The constant treatment effect assumption seems hard to swallow in many applications, especially when we think the treatment is chosen by agents based on differing gains from treatment.

To see the problem, consider a relaxation of the original linear model that allows for heterogeneity:

$$Y_i(d) = B_i d + U_i.$$

Note that $Y_i(d_1) - Y_i(d_0) = B_i(d_1 - d_0)$. The original IV estimator is

$$\beta_{\text{IV}} = \frac{\text{Cov}[Y_i, Z_i]}{\text{Cov}[D_i, Z_i]} = \frac{\text{Cov}[B_i D_i, Z_i]}{\text{Cov}[D_i, Z_i]} = \mathbb{E}\left[B_i \frac{D_i(Z_i - \mathbb{E}[Z_i])}{\text{Cov}[D_i, Z_i]}\right].$$

This is a weighted average of B_i's, but the weights can be negative for some i. In particular, it is possible to have $B_i > 0$ with probability one, but $\beta_{\text{IV}} < 0$.

Two major ways to address the heterogeneity issue:

- Forward engineering (Heckman): develop a new model and a new estimator.
- Reverse engineering (Imbens & Angrist): reinterpret the linear IV estimator with additional assumptions.

9.1 The LATE

We will first develop a simple model of treatment selection. Consider the binary case $D_i, Z_i \in \{0, 1\}$, with potential treatment choices

$$D_i = (1 - Z_i)D_i(0) + Z_i D_i(1).$$

We will make the following assumptions:

Assumption 9.1.

- **Full exogeneity**: $(Y_i(0), Y_i(1), D_i(0), D_i(1))$ is independent of Z_i.
- **Monotonicity**: $\mathbb{P}[D_i(1) \geq D_i(0)] = 1$. Equivalently, there are no **defiers**: $\mathbb{P}[D_i(0) = 1, D_i(0) = 0] = 0$.

Note that *full* exogeneity is stronger than our previous exogeneity assumption of $(Y_i(0), Y_i(1))$ being independent of Z_i, but it allows us to identify $\mathbb{E}[D_i(1) - D_i(0)]$ through the random assignment argument:

$$\mathbb{E}[Y_i|Z_i = z] = \mathbb{E}[Y_i(0)] + \mathbb{E}[D_i(z)B_i], \quad B_i := Y_i(1) - Y_i(0).$$

Thus the reduced form coefficient is, by the saturated regression logic,

$$\begin{aligned}
\frac{\text{Cov}[Y_i, Z_i]}{\text{Var}[Z_i]} &= \mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0] \\
&= \mathbb{E}[Y_i(0) + D_i(1)B_i|Z_i = 1] - \mathbb{E}[Y_i(0) + D_i(0)B_i|Z_i = 0] \\
&= \mathbb{E}[(D_i(1) - D_i(0))B_i] \\
&= \mathbb{E}[B_i|D_i(0) = 0, D_i(1) = 1]\,\mathbb{P}[D_i(0) = 0, D_i(1) = 1],
\end{aligned}$$

where the last equality comes from monotonicity. On the other hand, the first stage coefficient is

$$\begin{aligned}\frac{\mathrm{Cov}[D_i, Z_i]}{\mathrm{Var}[Z_i]} &= \mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0] \\ &= \mathbb{E}[D_i(1) - D_i(0)] = \mathbb{P}[D_i(0) = 0, D_i(1) = 1].\end{aligned}$$

Thus, the IV estimand

$$\beta_{\text{IV}} = \mathbb{E}[B_i|D_i(0) = 0, D_i(1) = 1]$$

can be interpreted as the **local average treatment effect** (LATE) for the **compliers**.

This result is intuitively appealing. After all, how can we learn about the treatment effect for the always-takers or never-takers if they never change their treatment status in response to the instrument? However, this also points to a potential externality validity issue. As always, how severe this problem is depends on the specific context.

9.2 Extending the LATE Interpretation

We have been working with the binary treatment and binary instrument case. Extensions to multivalued Z_i, multivalued D_i, or covariates W_i can be done, but they all come with caveats on interpretation and/or need extra assumptions.

Consider first the case of a multivalued instrument $Z_i \in \{z_0, \dots, z_k\}$. We impose the following monotonicity assumption:

Assumption 9.2.

$$\mathbb{P}\left[D_i(z_0) \le D_i(z_1) \le \cdots \le D_i(z_k)\right] = 1.$$

There are now multiple complier groups, one for each incremental change in Z_i. Their shares can be identified using contrasts in $p(z) \coloneqq \mathbb{P}[D_i = 1|Z_i = z]$.

$D_i(z_0)$	$D_i(z_1)$	$D_i(z_2)$	$D_i(z_3)$	g	$\mathbb{P}[G_i = g]$ is identified as
1	1	1	1	always treated	$p(z_0)$
0	1	1	1	$\textsc{cp}_1$, complier 1	$p(z_1) - p(z_0)$
0	0	1	1	$\textsc{cp}_2$, complier 2	$p(z_2) - p(z_1)$
0	0	0	1	$\textsc{cp}_3$, complier 3	$p(z_3) - p(z_2)$
0	0	0	0	never treated	$1 - p(z_3)$

To estimate treatment effects, we could just focus on any two values of Z_i:

$$\text{Wald estimand} \coloneqq \frac{\mathbb{E}[Y_i|Z_i = z_j] - \mathbb{E}[Y_i|Z_i = z_{j-1}]}{\mathbb{E}[D_i|Z_i = z_j] - \mathbb{E}[D_i|Z_i = z_{j-1}]} = \mathbb{E}[B_i|G_i = \textsc{cp}_j] = \textsc{late}_j.$$

In this binary contrast, the Wald estimand is just another expression for β_{IV}.

We can also fully integrate Z_i into our regression. Consider any function $\zeta(Z_i)$ (identity, or including interactions or higher order terms, etc.). We have, with some annoying algebra,

$$\beta_{\text{IV}}(\zeta) \coloneqq \frac{\mathrm{Cov}[Y_i, \zeta(Z_i)]}{\mathrm{Cov}[D_i, \zeta(Z_i)]} = \sum_{j=1}^{k} \left[\frac{\mathbb{P}[G_i = \textsc{cp}_j]\,\mathrm{Cov}[\zeta(Z_i), \mathbb{1}\left\{Z_i \in \{z_\ell\}_{\ell=j}^{k}\right\}]}{\mathrm{Cov}[D_i, \zeta(Z_i)]}\right] \cdot \textsc{late}_j,$$

which is a weighted average of the LATE$_j$'s, with weights positive if $\zeta(Z_i)$ is increasing in Z_i. However, different linear IV specifications can give different weights, and thus different estimates.

10 IV Summary

10.1 Linear IV

Assumption 10.1.

(i) Exclusion: $Y_i(d, z) = Y_i(d)$.

(ii) Exogeneity: $Y_i(d) \perp\!\!\!\perp Z_i | W_i$.

(iii) Linear functional form:

$$Y_i(d) = g(d, W_i)'\beta + U_i = X_i'\beta + U_i, \qquad \mathbb{E}[U_i | W_i].$$

(iv) Relevance: $\mathrm{Cov}[D_i, Z_i] \neq 0$ or $\mathbb{E}[cF_iX_i']$ has full column rank for some c, with $F_i = [Z_i', W_i']'$.

(v) Consequence: $\mathbb{E}[U_i | Z_i, W_i] = 0$.

Estimators:

$$\beta_1 = \frac{\mathrm{Cov}[Y_i, Z_i]}{\mathrm{Cov}[D_i, Z_i]},$$
$$\beta = \mathbb{E}[cF_iX_i']^{-1}\,\mathbb{E}[cF_iY_i].$$

$$c_{\text{TSLS}} = \mathbb{E}[X_iF_i']\,\mathbb{E}[F_iF_i']^{-1}, \quad c_{\text{OGMM}} = \mathbb{E}[X_iF_i']\,\mathbb{E}[U_i^2F_iF_i']^{-1}.$$

Variance:

$$\Sigma_{\text{IV}}(c) = \mathbb{E}[cF_iX_i']^{-1}cE[U_i^2F_iF_i']c'E[F_iX_i'c']^{-1}.$$

10.1.1 Weak Instrument Diagnostics

$$T_{\text{FSFS}} \coloneqq \frac{1}{d_z}\left\|(V_{\text{HOM}}/n)^{-1/2}C\right\|^2 \xrightarrow{\text{p}} \frac{1}{d_z}\left\|\frac{\mathbb{E}[Z_iZ_i']^{1/2}\gamma}{\sigma_V} + R\right\|^2 \eqqcolon \frac{1}{d_z}\|\mu + R\|^2,$$

where R is a standard normal and μ^2 is the concentration parameter.

10.1.2 The Anderson-Rubin Test

Test $H_0 : \beta_1 = \beta_{1,0}$. Define $U_i(\beta_{1,0}) \coloneqq Y_i - X_{i1}'\beta_{1,0}$. Under H_0, $\mathbb{L}[U_i(\beta_{1,0})|F_i] = 0$, which can be tested by a F test.

10.2 The LATE Model

Assumption 10.2 (Binary treatment and binary instrument, no covariates).

(i) Heterogeneous treatment effects:

$$Y_i(d) = B_id + U_i.$$

(ii) Full exogeneity: $Y_i(1), Y_i(0), D_i(1), D_i(0) \perp\!\!\!\perp Z_i$.

(iii) Monotonicity: $D_i(1) \geq D_i(0)$ almost surely.

Can be generalized to (ordered) multivalued instruments. The Wald estimators give the LATEs:

$$\frac{\mathbb{E}[Y_i|Z_i = z_j] - \mathbb{E}[Y_i|Z_i = z_{j-1}]}{\mathbb{E}[D_i|Z_i = z_j] - \mathbb{E}[D_i|Z_i = z_{j-1}]} = \mathbb{E}[Y_i(1) - Y_i(0)|D_i(z_j) = 1, D_i(z_{j-1}) = 0].$$